Probability Theory

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Contents

1 Probability Theory (機率論)

I Basic definitions

i (Random) Experiment or Trial (（隨機）試驗)

A (random) experiment or trial is a test that has repeatability, that is, the test can be repeated under the same conditions, and randomness, that is, the results of each test may be different.

ii (Elementary) outcome and sample space (樣本空間)

The sample space of an experiment is the finest-grain, mutually exclusive, collectively exhaustive set of all possible (elementary) outcomes. For example, the sample space for flipping a coin is $\{\text{head}, \text{tail}\}$.

The singleton of an outcome is sometimes denoted the same as the outcome.

iii Event (事件)

A subset of the sample space. For example, the event of rolling a die and getting an even number is $\{2, 4, 6\}$.

iv Event space

The set of all events of an experiment.

v Probability (機率)

The likelihood of an event occuring, which is a real number in $[0, 1]$. The closer the probability is to 1, the more likely the event is to occur. The probability of an event A occuring is denoted as $P(A)$, $\mathbb{P}(A)$, $P[A]$, or $\mathbb{P}[A]$.

vi Impossible event (空事件)

An event with zero probability.

vii Sure event (全事件)

An event with one probability. If A is a sure event, the trial is said to be almost surely (a.s.) A.

viii Sum event (和事件)

The sum event of event A and event B is $A \cup B$, also denoted as $A + B$.

ix Product event (積事件)

The product event of event A and event B is $A \cap B$, also denoted as AB or A, B.

x Complement event (餘事件)

The complement event of event A in sample space S is $S \setminus A$, denoted as A', A^c, or A^C.

xi Probability space (機率空間)

A probability space $(\Omega, \mathcal{F}, P)$ is a measure space of which Ω is the sample space of an experiment, $\mathcal{F}$ is the event space of the experiment, and P is the probability measure with unit measure axiom $P(\Omega) = 1$ that maps each event to the probability of it. The measurable space (sample space and event space tuple) is called state space.

xii Probability axioms or Kolmogorov axioms

1. Nonnegativity:
$$\forall E \in \mathcal{F}: \ P(E) \geq 0.$$

2. Unit measure or normalization:
$$P(\Omega) = 1.$$

3. Countable additivity or σ-additivity: For all indexed family $(E_k)_{k \in \mathbb{N}}$ of pairwise disjoint sets in $\mathcal{F}$,
$$P\left(\bigcup_{k \in \mathbb{N}} E_k\right) = \sum_{k=1}^{\infty} P(E_k).$$

Countable additivity implies $P(\varnothing) = 0$.

Proof. By applying for $(\varnothing)_{k \in \mathbb{N}}$,
$$P(\varnothing) = \sum_{k=1}^{\infty} P(\varnothing)$$
□

Countable additivity implies finite additivity: For all indexed family $(E_k)_{k \in \mathbb{N} \cap [1,n]}$ of pairwise disjoint sets in $\mathcal{F}$,
$$P\left(\bigcup_{k=1}^{n} E_k\right) = \sum_{k=1}^{n} P(E_k).$$

Proof. Let $(F_k)_{k \in \mathbb{N}}$ be defined as $F_k = E_k$ for $k \in [1, n]$ and $F_k = \varnothing$ for $k \in (n, \infty)$ and countable additivity. □

xiii Conditional probability (條件機率)

The probability of another event A occurring given that a certain event B has occurred, denoted as $P(A|B)$, is defined as
$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

xiv Partition (分割)

A partition is a partition of sample space that is a subset of the event space.

xv Classical probability (古典機率) or equally likely outcomes

If the number of all possible outcomes in the sample space S of an experiment is finite, and the chance of each outcome occurring is equal, then the probability of an event A occurring can be calculated by:

$$P(A) = \frac{|A|}{|S|}.$$

xvi Mutually exclusive events (互斥事件)

Two or more events $A_1, A_2, \dots, A_n$ are mutually exclusive events iff:

$$\forall i, j \in \mathbb{N}_{\leq n} \wedge i \neq j : \; P(A_i \cap A_j) = 0.$$

xvii (Mutually) independent events (獨立事件)

Two or more events $A_1, A_2, \dots, A_n$ are (mutually) independent events iff:

$$\forall J \subseteq \mathbb{N}_{\leq n} \wedge J \neq \varnothing : \; P\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} P\left(A_j\right).$$

xviii Sequential experiment

A sequential experiment is an experiment that

- Consists of two or more experiments called subexperiments or stages.
- Has outcomes at each stage, which may depend on the outcome of the previous stages.
- The overall outcome of the experiment is an ordered tuple of stage outcomes

$$\omega = (\omega_1, \omega_2, \dots, \omega_n).$$

xix Tree diagram

Tree diagram is often used to represent the possible outcomes, in which the root is the beginning of the experiment, each branch represents a possible stage outcome, and a path from the root to a leaf represents a complete outcome of the sequential experiment. Conditional probabilities can be assigned to branches to compute the probability of each path as the product of the conditional probabilities of all branches in the path.

xx Independent trials

Independent trials are identical subexperiments in a sequential experiment. The probability space of all the subexperiments are identical and independent of the outcomes of previous subexperiments. Let an event of the sequential experiment be

$$\omega = (\omega_1, \omega_2, \dots, \omega_n),$$

then

$$P(\omega) = \prod_{i=1}^{n} P(\omega_i).$$

xxi Reliability theory

A system is in series if all components work iff the system work.

A system is in parallel if at least one component works iff the system work.

II Counting principle (計數原理) or fundamental principle of counting (基本計數原理)

i Complement principle

$$P(A') = 1 - P(A).$$

ii De Morgan's laws

$$P((A \cup B)') = P(A' \cap B').$$

$$P((A \cap B)') = P(A' \cup B').$$

iii Principle of inclusion-exclusion (PIE) or inclusion–exclusion principle (排容原理/取捨原理)

For event space $\mathcal{F}$ and indexed family $(A_i)_{i=1}^n \subseteq \mathcal{F}$ with $n \in \mathbb{N}$,

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{J \subseteq \mathbb{N}_{\leq n} \wedge J \neq \varnothing} (-1)^{|J|+1} P\left(\bigcap_{j \in J} A_j\right).$$

iv Subset law

$$A \subseteq B \implies P(A) \leq P(B).$$

v Union bound

$$P(\bigcup_{i=1}^n A_i) \leq \sum_{i=1}^n P(A_i).$$

III Independence theorems

i Product rule

If events A and B are independent, $P(A) \cdot P(B) = P(A \cap B)$.

ii Complement independence

If $A_1, A_2, \dots, A_n$ are independent, then they, with arbitrary ones replaced with their complement, are still independent.

iii Pairwise and mutal

Pairwise independence is implied by but do not imply mutual independence.

iv Multinominal distribution

Suppose a subexperiment has sample space $S = \{s_0, s_1, \dots, s_{m-1}\}$ with $P(s_i) = p_i$. For $n = \sum_{i=0}^{m-1} n_i$ independent trials, the probability of n_i occurrences of s_i is

$$\binom{n}{n_0, n_1, \dots, n_{m-1}} \prod_{i=0}^{m-1} p_i^{\,n_i}.$$

v Components in series

Suppose a system W consists of components $W_1, W_2, \dots, W_n$ in series with probability to work $p_1, p_2, \dots, p_n$, then the probability for the system to work is

$$P(W) = \prod_{i=1}^{n} p_i.$$

vi Components in parallel

Suppose a system W consists of components $W_1, W_2, \dots, W_n$ in parallel with probability to work $p_1, p_2, \dots, p_n$, then the probability for the system to work is

$$P(W) = 1 - \prod_{i=1}^{n} (1 - p_i).$$

IV Random variable (RV) (隨機變數)

i Random variable

For a probability spaces $(\Omega_s, \mathcal{F}_s, P_s)$ and a measurable space $(\Omega_X, \mathcal{F}_X)$, a random variable X is a measurable function $X : \Omega_s \to \Omega_X$ such that for each $E \subseteq \mathcal{F}_X$, the preimage $X^{-1}(E) \in \mathcal{F}_s$, and the induced probability spaces $(\Omega_X, \mathcal{F}_X, P_X)$ of target sample space Ω_X, target event space, and (pushforward) probability measure $P_X(E) \coloneqq P_s(X^{-1}(E))$. It is often the case that $\mathcal{F}_X$ is the Borel σ-algebra of Ω_X.

ii Real-valued random variable

A random variable is a real-valued or real random variable if $\Omega_X \subseteq \mathbb{R}$ (equipped with subspace topology) and $\mathcal{F}_X$ is the Borel σ-algebra of Ω_X. Random variable often refers to real-valued random variable.

iii Cumulative distribution function (CDF)

The cumulative distribution function (CDF) of a real-valued random valuable X is defined as

$$F_X(x) \coloneqq P(X \le x) = \int_{\Omega_X} \mathbf{1}_{(-\infty, x]}(t) \, \mathrm{d}P_X(t) \, .$$

iv Complementary cumulative distribution function (CCDF), tail distribution function, or exceedance

The complementary cumulative distribution function (CCDF) of a real-valued random valuable X is defined as

$$\bar{F}_X(x) = 1 - F_X(x).$$

v CDF range

$$0 \leq F_X(x) \leq 1$$

vi CDF difference theorem

For any $a \geq b$,

$$F_X(b) - F_X(a) = P(X \in (a, b]).$$

vii Non-decreasing property of CDF

A CDF is non-decreasing.

Proof. For $x < y$, $\{X | X \leq x\} \subseteq \{X | X \leq y$, thus $P(X \leq x) \leq P(X \leq y)$. □

viii Limit at negative infinity of CDF

$$\lim_{x \to -\infty} F_X(x) = 0.$$

Proof.

$$x_1 > x_2 \implies \{X \leq x_1\} \supseteq \{X \leq x_2\}$$
$$\bigcap_{x \in \mathbb{R}} \{X \leq x\} = \varnothing$$

By continuity of measure from above,

$$\lim_{x \to -\infty} P(X \leq x) = P(\varnothing) = 0$$

□

ix Limit at positive infinity of CDF

$$\lim_{x \to \infty} F_x(x) = 1.$$

Proof.

$$x_1 < x_2 \implies \{X \leq x_1\} \subseteq \{X \leq x_2\}$$
$$\bigcup_{x \in \mathbb{R}} \{X \leq x\} = \Omega_X$$

By continuity of measure from below,

$$\lim_{x \to \infty} P(X \leq x) = P(\Omega_X) = 1$$

□

x Right-continuity of CDF

A CDF is right-continuous.

Proof.

$$h_1 > h_2 \implies \{X \le x + h_1\} \supseteq \{X \le x + h_2\}$$

$$\bigcap_{h>0} \{X \le x + h\} = \{X \le x\}$$

By continuity of measure from above,

$$\lim_{h \to 0^+} P(X \le x + h) = P\left(\bigcap_{h>0} \{X \le x + h\}\right) = P(X \le x)$$

□

xi Converse property of CDF

If a function $F : \mathbb{R} \to [0, 1]$ is non-decreasing, has limits 0 at $-\infty$ and 1 at ∞, and is right-continuous, then there exists a real-valued random variable X such that $F_X = F$.

Proof. Let $\Omega_X = (0, 1)$, $\mathcal{F} = \mathcal{B}(\Omega)$, and P_X be Lebesgue measure.

Define

$$X(u) \coloneqq \inf\{x \in \mathbb{R} \mid F(x) \ge u\}, \quad u \in (0, 1)$$

$$F_X(x) = F(x)$$

□

xii Quantile

Quantile $Q_X(p)$ is the inverse of CDF, that is,

$$P(X \le Q_X(p)) = p.$$

xiii Distribution or law

The distribution or law of X is a complete model that uniquely determines P_X, or equivalently, a complete model that uniquely determines F_X.

xiv Discrete random variable

A discrete random variable is a real-valued random variable such that there exists a countable subset A of sample space such that $P(A) = 1$.

xv Continuous random variable

A continuous random variable is a real-valued random variable with continuous CDF.

xvi Properties of continuous random variable

For all $\omega \in \Omega_X$,

$$P(X = \omega) = 0.$$

xvii Absolutely continuous random variable

A continuous random variable is a real-valued random variable with absolutely continuous CDF. Absolutely continuous random variable is often mistakenly written as continuous random variable, even in many textbooks.

xviii Mixed random variable

A mixed random variable is a real-valued random variable X whose CDF F_X is

$$pF_Y + (1-p)F_Z$$

for some $p \in (0,1)$, discrete random variable Y, and absolutely continuous random variable Z.

xix Singular random variable

A singular random variable is a real-valued random variable that is neither discrete random variable, continuous random variable, nor mixed random variable.

xx Simple random variable

A simple random variable is a discrete random variable X such that there exist a finite partition $\{A_1, \dots, A_n\}$ of source sample space Ω such that $X = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$ where a_i are real numbers and $\mathbf{1}_{A_i}$ is the indicator function of A_i, that is,

$$\mathbf{1}_{A_i}(\omega) = \begin{cases} 1, & \omega \in A_i \\ 0, & \omega \notin A_i \end{cases}.$$

xxi Decomposition

For the CDF F_X of any real-valued random variable X, there exist $p \in [0,1]$, $q \in [0,1]$, discrete random variable Y, absolutely continuous random variable Z, and singular random variable W such that

$$F_X = pF_Y + qF_z + (1-p-q)F_W.$$

xxii Equality in distribution or law

Two real-valued random variables are called equal in distribution or law if they have the same pushforward probability measure.

Two real-valued random variables are equal in distribution iff they have the same CDF.

Proof. By measures agree on generating π-system agree theorem. □

xxiii Infinitely often (i.o.)

For a sequence $(A_n)_{n\in\mathbb{Z}_{\geq k}}$ event for some $k \in \mathbb{Z}$,

$$A_n \text{ i.o. } := \bigcap_{m=k}^{\infty} \bigcup_{n\geq m} A_n.$$

xxiv Almost sure (a.s.) convergence

A sequence $(X_n)_{n\in\mathbb{Z}_{\geq k}}$ of random variables for some $k \in \mathbb{Z}$ almost surely coverges to a random variable X, denoted as

$$X_n \xrightarrow{a.s.} X \text{ as } n \to \infty,$$

$$X_n \xrightarrow{a.s.} X \text{ when } n \to \infty,$$

if

$$P(\lim_{n\to\infty} X_n = X) = 1.$$

xxv Convergence in probability

A sequence $(X_n)_{n\in\mathbb{Z}_{\geq k}}$ of random variables for some $k \in \mathbb{Z}$ of random variables coverges in probability to a random variable X, denoted as

$$X_n \xrightarrow{P} X \text{ as } n \to \infty,$$

$$X_n \xrightarrow{p} X \text{ as } n \to \infty,$$

$$X_n \xrightarrow{P} X \text{ when } n \to \infty,$$

$$X_n \xrightarrow{p} X \text{ when } n \to \infty,$$

if for all $\varepsilon > 0$,

$$\lim_{n\to\infty} P(\left|X_n - X\right| > \varepsilon) = 0.$$

xxvi Convergence in distribution or weak convergence

A sequence $(X_n)_{n\in\mathbb{Z}_{\geq k}}$ of random variables for some $k \in \mathbb{Z}$ of random variables coverges in distribution or weakly converges to a random variable X, denoted as

$$X_n \xrightarrow{d} X \text{ as } n \to \infty,$$

$$X_n \xrightarrow{\mathcal{D}} X \text{ as } n \to \infty,$$

$$X_n \xrightarrow{d} X \text{ when } n \to \infty,$$

$$X_n \xrightarrow{\mathcal{D}} X \text{ when } n \to \infty,$$

if for all x such that $F_X(x)$ is continuous at x,

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x).$$

xxvii Almost sure convergence implies convergence in probability

For a sequence $(X_n)_{n\in\mathbb{Z}\geq k}$ of random variables for some $k \in \mathbb{Z}$ of random variables, if

$$X_n \xrightarrow{a.s.} X \text{ as } n \to \infty,$$

then

$$X_n \xrightarrow{P} X \text{ as } n \to \infty.$$

Proof. For any fixed $\varepsilon > 0$. Define

$$A_n = \{\omega : \left|X_n(\omega) - X(\omega)\right| > \varepsilon\}.$$

Almost sure convergence implies that for almost every ω, there exists $N(\omega) \in \mathbb{N}$ such that

$$n \geq N(\omega) \implies \left|X_n(\omega) - X(\omega)\right| \leq \varepsilon.$$

Thus for almost every ω, there exists $N(\omega) \in \mathbb{N}$ such that

$$n \geq N(\omega) \implies \omega \notin A_n.$$

Thus,

$$P(\bigcap_{m=1}^{\infty} \bigcup_{n\geq m} A_n) = 0.$$

Observe that for all n,

$$P(A_n) \leq P(\bigcup_{m\geq n} A_m)$$

Let

$$B_n := \bigcup_{m\geq n} A_m,$$

which is a decreasing sequence

$$B_k \supseteq B_{k+1} \supseteq \dots$$

and

$$\bigcap_{n=k}^{\infty} B_n = \bigcap_{m=1}^{\infty} \bigcup_{n\geq m} A_n$$

Thus by continuity of probability measure from above,

$$\lim_{n\to\infty} P(B_n) = P(\bigcap_{m=1}^{\infty} \bigcup_{n\geq m} A_n) = 0$$

Since

$$P(A_n) \leq P(B_n),$$

we have

$$\lim_{n\to\infty} P(A_n) = 0.$$

That is,

$$\lim_{n\to\infty} P(\left|X_n - X\right| > \varepsilon) = 0.$$

□

xxviii Convergence in probability implies convergence in distribution

For a sequence $(X_n)_{n\in\mathbb{Z}\geq k}$ of random variables for some $k \in \mathbb{Z}$ of random variables, if

$$X_n \xrightarrow{P} X \text{ as } n \to \infty,$$

then

$$X_n \xrightarrow{d} X \text{ as } n \to \infty.$$

Proof. For any fixed x such that $F_X(x)$ is continuous at x, for any fixed $\varepsilon > 0$,

$$\{X \leq x - \varepsilon\} \cap \{\left|X_n - X\right| \leq \varepsilon\} \subseteq \{X_n \leq X\} \subseteq \{X \leq x + \varepsilon\} \cup \{\left|X_n - X\right| > \varepsilon\}.$$

$$P(X \leq x - \varepsilon) - P(\left|X_n - X\right| > \varepsilon) \leq P(X_n \leq X) \leq P(X \leq x + \varepsilon) + P(\left|X_n - X\right| > \varepsilon).$$

Since

$$\lim_{n\to\infty} P(\left|X_n - X\right| > \varepsilon) = 0,$$

$$\liminf_{n\to\infty} P(X_n \leq X) \geq P(X \leq x - \varepsilon)$$

and

$$\limsup_{n\to\infty} P(X_n \leq X) \leq P(X \leq x + \varepsilon).$$

Since $F_X(x)$ is continuous at x,

$$\lim_{\varepsilon\to 0} P(X \leq x - \varepsilon) = \lim_{\varepsilon\to 0} P(X \leq x + \varepsilon) = P(X \leq x).$$

By squeeze theorem,

$$\lim_{n\to\infty} P(X_n \leq X) = P(X \leq x).$$

□

xxix Convergence in distribution to a constant implies convergence in probability to it

For a sequence $(X_n)_{n\in\mathbb{Z}\geq k}$ of random variables for some $k \in \mathbb{Z}$ of random variables and a constant C, if

$$X_n \xrightarrow{d} C \text{ as } n \to \infty,$$

then

$$X_n \xrightarrow{P} C \text{ as } n \to \infty.$$

Proof.

$$F_C(x) = \begin{cases} 0, & x \leq c, \\ 1, & x > c. \end{cases}$$

Convergence in distribution means

$$\forall x < c: \ P(X_n \leq x) \to 0$$

$$\forall x > c: \ P(X_n \leq x) \to 1$$

$$\lim_{n\to\infty} P(X_n \le c - \varepsilon) = 0.$$

$$\lim_{n\to\infty} P(X_n \le c + \varepsilon) = 1.$$

$$\lim_{n\to\infty} P(X_n > c + \varepsilon) = \lim_{n\to\infty} 1 - P(X_n \le c + \varepsilon) = 0.$$

□

V Discrete Random Variables

i Probability mass (function) (PMF) or probability function/distribution for absolutely continuous random variable

For a discrete random variable X with target sample space $\Omega_X = \{a_1, a_2, \dots\}$, the probability mass function of X, denoted as p_X or P_X, is defined as

$$p_X(a_i) = P(X = a_i)$$

for each $a_i \in \Omega_X$.

That the PMF of X is p is denoted as $X \sim p$.

ii PMF determines CDF

For a discrete random variable X with PMF p_X, the CDF F_X can be expressed in terms of the PMF as

$$F_X(x) = \sum_{a_i \le x} p_X(a_i)$$

for all $x \in \mathbb{R}$.

iii Support

The support Supp(X) of a discrete random variable X is the support of its PMF, which is a finite or countable set.

iv CDF as combination of indicator functions

A real-valued random variable X is a discrete random variable iff its CDF F_X is such that there exists a finite or countable set $A \subseteq \mathbb{R} \times (0, 1]$ such that

$$\sum_{(a,p)\in A} p = 1$$

and

$$F_X(x) = \sum_{(a,p)\in A} p\mathbf{1}_{[a,\infty)}(x).$$

If so, $\pi_1(A) = $ Supp(X), and for any $(a, p) \in A$, $p = p_X(a)$.

v Properties

$$p_X(a_i) \geq 0, \quad \forall a_i \in \Omega_X.$$

$$\sum_{a_i \in \Omega_X} p_X(a_i) = 1.$$

$$P(X \in A) = \sum_{a_i \in A} p_X(a_i), \quad \forall A \in \mathcal{F}_X.$$

$$\frac{\mathrm{d}F_X}{\mathrm{d}x} = 0 \quad \text{a.e..}$$

vi Probability simplex

Given a sample space $(x_0, x_1, x_2, \ldots, x_k)$, each point $(p_0, p_1, \ldots, p_k)$ in the probability k-simplex

$$\{(p_0, p_1, \ldots, p_k) \mid \sum_{i=0}^{k} p_i = 1\}$$

corresponds to the PMF p defined as

$$p(x_i) = p_i, \quad i \in [0, k] \cap \mathbb{Z}.$$

VI Absolutely Continuous Random Variables

i Probability density (function) (PDF), density, or probability function/distribution for absolutely continuous random variable

For an absolutely continuous random variable X, the probability density function of X, denoted as f_X, is defined as

$$f_X(x) = \frac{\mathrm{d}}{\mathrm{d}x} F_X(x),$$

and for those points at which F_X is not differentiable, arbitrary real number value.

Since F_X is absolutely continuous, it is differentiable a.e., and for all f_X, g_X such that

$$\int_{-\infty}^{x} f_X(t) \,\mathrm{d}t = \int_{-\infty}^{x} g_X(t) \,\mathrm{d}t = F_X(x),$$

we have

$$f_X(x) = g_X(x)$$

a.e., and

$$f_X(x) = F_X'(x)$$

a.e. Equality or inequality involving PDF means a.e.

That the PDF of X is p is denoted as $X \sim p$.

ii PDF determines CDF

For an absolutely continuous random variable X with PDF f_X, the CDF F_X can be expressed in terms of the PMF as

$$F_X(x) = \int_{-\infty}^{x} f_x(t) \,\mathrm{d}t$$

for all $x \in \mathbb{R}$.

iii Support

The support $\mathrm{Supp}(X)$ of an absolutely continuous random variable X is the support of its PDF.

iv Properties

$$f_X(x) \geq 0.$$

$$\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1.$$

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,\mathrm{d}x\,.$$

VII As Schwartz distribution

i CDF as a distribution

F_X is bounded and monotone, thus in L^1_{loc}. So it induces a distribution T_X:

$$\langle T_X, \varphi \rangle = \int_{-\infty}^{\infty} F_X(x)\varphi(x)\,\mathrm{d}x$$

ii PDF and probability measure as a distribution

PDF as a distribution $t_X = T_F'$, that is,

$$\langle t_X, \varphi \rangle = -\int_{-\infty}^{\infty} F_X(x)\varphi'(x)\,\mathrm{d}x$$

It is the same as the pushforward probability measure P_X of X as a distribution:

$$-\int_{-\infty}^{\infty} F_X(x)\varphi'(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} \varphi(x)\,\mathrm{d}P_X(x)\,.$$

iii Absolutely continuous case

t_X is the same as f_X as a distribution.

iv Discrete case

$$t_X = \sum_{\omega \in \Omega_X} p_X(\omega)\delta_\omega$$

v Mixed case

Suppose

$$pF_Y + (1-p)F_Z$$

for some $p \in (0,1)$, discrete random variable Y, and absolutely continuous random variable Z.

PDF as a distribution t_X of X is

$$t_X = pt_Y + (1-p)t_Z.$$

vi Properties of impulse

Suppose

$$F_X = pF_Y + qF_z + (1-p-q)F_W$$

for some $p \in [0,1]$, $q \in [0,1]$, discrete random variable Y, absolutely continuous random variable Z, and singular random variable W. The following statements are equivalent:

$$P(X = a) = r$$

$$F_X(a^+) - F_X(a^-) = \frac{r}{p} \wedge p > 0$$

$$P(Y = a) = pr$$

$$t_X = r\delta_a + t_V$$

for some real-valued random variable V with $P(V = a) = 0$.

VIII Conditional Probability

i Conditional probability space

For a probability space $(\Omega, \mathcal{F}, P)$ and a fixed event $B \in \mathcal{F}$ with $P(B) > 0$, the function $P(\cdot|B) : \mathcal{F} \to [0,1]$ defined by $A \mapsto P(A|B)$ is a probability measure on the same measurable space $(\Omega, \mathcal{F})$.

ii Conditional random variable

For a probability space $(\Omega_s, \mathcal{F}_s, P_s)$, a measurable space $(\Omega_X, \mathcal{F}_X)$, a random variable $X : \Omega_s \to \Omega_X$ with pushforward probability measure P_X, and a fixed event $A \in \mathcal{F}_X$ with $P_X(A) > 0$, the condition random variable $X|A$ is a randm variable over $(\Omega_s, \mathcal{F}_s, P_s(\cdot|X^{-1}(A)))$ with pushforward probability measure $P_X(\cdot|A)$.

iii Conditional CDF

For a real-valued random variable X and an event $A \in \mathcal{F}_X$ such that $P(A) > 0$, the conditional CDF of X given A follows

$$F_{X|A}(x) = P(X \le x|A) = \frac{P(X \le x, A)}{P(A)}, \quad \forall x \in \mathbb{R}.$$

iv Conditional PMF of discrete random variable

For a discrete random variable X and an event $A \in \mathcal{F}_X$ such that $P(A) > 0$, the conditional PMF of X given A follows

$$p_{X|A}(x) = P(X = x|A) = \frac{P(X = x, A)}{P(A)} = \frac{P(X = x) \cdot \mathbf{1}_A(x)}{P(A)}, \quad \forall x \in \mathbb{R}.$$

v Conditional PDF of absolutely continuous random variable

For an absolutely continuous random variable X and an event $A \in \mathcal{F}_X$ such that $P(A) > 0$, the conditional PDF of X given A follows

$$f_{X|A}(x) = \frac{\mathrm{d}}{\mathrm{d}x} F_{x|A}(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{P(X \le x, A)}{P(A)}\right) = \frac{f_X(x) \cdot \mathbf{1}_A(x)}{P(A)}, \quad \forall x \in \mathbb{R}.$$

vi Regular conditional probability

Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $T : \Omega \to E$ be a random variable that is a measurable function to measurable space $(E, \mathcal{E})$ with pushforwad probability measure $P \circ T^{-1}$. A regular conditional probability or transition probability is defined as a function $\nu : E \times \mathcal{F} \to [0,1]$, where:

- For every $x \in E$, $\nu(x, \cdot)$ is a probability measure on $\mathcal{F}$,
- For every $A \in \mathcal{F}$, $\nu(\cdot, A)$ is $\mathcal{E}$-measurable, and
- For every $A \in \mathcal{F}$ and $B \in \mathcal{E}$,

$$P(A \cap T^{-1}(B)) = \int_B \nu(x, A)\, \mathrm{d}(P \circ T^{-1})(x)\,.$$

$\nu(x, A)$ is typically denoted as $P(A|T = x)$.

For any teo regular conditional probabilities ν_1, ν_2, for every $A \in \mathcal{F}$,

$$\nu_1(x, A) = \nu_2(x, A)$$

for $P \circ T^{-1}$-almost every $x \in E$.

IX Conditional probability laws

i Law of multiplication

For sample space Ω and n events $A_1, A_2, \dots, A_n$, define $B_i = \bigcap_{j=1}^{i-1} A_j$ for $i \in \mathbb{N}_{\geq 2, \leq n+1}$ and $B_1 = \Omega$. Then,

$$P(\bigcap_{i=1}^{n} A_i) = \begin{cases} \prod_{i=1}^{n} P(A_i|B_i), & \forall i \in \mathbb{N}_{\leq n} :\ P(B_i) > 0 \\ 0, & \exists i \in \mathbb{N}_{\leq n} :\ P(B_i) = 0 \end{cases}.$$

Proof.

$$\prod_{i=1}^{n} P(A_i|B_i) = \prod_{i=1}^{n} \frac{P(B_{i+1})}{P(B_i)} = \frac{P(B_{n+1})}{P(B_1)} = \frac{P(\bigcap_{i=1}^{n} A_i)}{1} = P(\bigcap_{i=1}^{n} A_i).$$

□

ii Law of total probability

Let $(A_i)_{i=1}^n \subseteq \mathcal{F}$ be a partition, and B be an event. Then

$$P(B) = \sum_{i=1}^{n} P(A_i \cap B) = \sum_{i=1}^{n} P(B|A_i)P(A_i).$$

Let X be a discrete random variable. Then

$$p_X(x) = \sum_{i=1}^{n} p_{X|A_i}(x)P(A_i).$$

Let X be an absolutely continuous random variable. Then

$$f_X(x) = \sum_{i=1}^{n} f_{X|A_i}(x) P(A_i).$$

iii Bayes' theorem (貝葉斯定理或貝氏定理)

Let $(A_i)_{i=1}^n \subseteq \mathcal{F}$ be a partition, and B be an event. Then,

$$P\left(A_k \mid B\right) = \frac{P\left(B|A_k\right) P(A_k)}{\sum_{i=1}^n P\left(B \mid A_i\right) P(A_i)}.$$

X Joint random variable, multivariate random variable, or random vector

i Definition

For d real-valued random variables $X_1, \dots, X_d$ with source sample spaces $\Omega_1, \dots, \Omega_d$ and source event spaces $\mathcal{F}_1, \dots, \mathcal{F}_d$, the multivariate random variable, joint random variable, or random vector $X = (X_1, \dots, X_d)$ has source sample space $\Omega = \prod_{i=1}^{d} \Omega_i$ and source event space $\mathcal{F} = \prod_{i=1}^{d} \mathcal{F}_i$ and is defined for all $\omega = (\omega_1, \dots, \omega_d) \in \Omega$ as

$$X(\omega) = (X_1(\omega_1), \dots, X_d(\omega_d)).$$

For $\mathcal{F}_1, \dots, \mathcal{F}_d = \mathcal{B}(\mathbb{R})$, $\mathcal{F} = \mathcal{B}(\mathbb{R}^d)$.

ii (Joint, multivariable, or multivariate) cumulative distribution function (CDF)

$$F_X(x_1, \dots, x_d) = P(X_1 \le x_1, \dots, X_d \le x_d),$$

where , denotes logical AND.

iii Joint CDF range

$$0 \le F_X(x_1, \dots, x_d) \le 1$$

iv Coordinate non-decreasing property of joint CDF

For any $(a_1, \dots, a_d)$ and $(b_1, \dots, b_d)$ with $a_i \le b_i$ for all $i \in \mathbb{Z} \cap [1, d]$,

$$F_X(a_1, \dots, a_d) \le F_X(b_1, \dots, b_d).$$

v Limit as any to negative infinity of joint CDF

For any $i \in \mathbb{Z} \cap [1, d]$,

$$\lim_{x_i \to -\infty} F_X(x_1, \dots, x_d) = 0.$$

vi Limit as all to positive infinity of CDF

$$\lim_{(x_1,\ldots,x_d)\to(\infty,\ldots,\infty)} F_X(x_1,\ldots,x_d) = 1.$$

vii Right-continuity of CDF

$$\lim_{x_1\to {a_1}^+,\ldots,x_d\to {a_d}^+} F_X(x_1,\ldots,x_d) = F_X(a_1,\ldots,a_d).$$

viii Rectangle increment

For any rectangle $R = \prod_{i=1}^{d}(a_i, b_i]$, $\Delta F_X(R)$ denotes $P(a_1 < X_1 \le b_1, \ldots, a_d < X_d \le b_d)$, that is,

$$\Delta F_X(R) = \sum_{(\varepsilon_1,\ldots,\varepsilon_d)\in\{0,1\}^d} (-1)^{d-\sum_{i=1}^d \varepsilon_i} F_X(a_1+\varepsilon_1(b_1-a_1),\ldots,a_d+\varepsilon_d(b_d-a_d)).$$

Proof. Define $A_i = X_i \le b_i$, $B_i = X_i \le a_i$, $A_i \supseteq B_i$. Then $A_i \setminus B_i = A_i \cap {B_i}^c = a_i < X_i \le b_i$. By inclusion-exclusion principle,

$$\begin{aligned}
\Delta F_X(R) =& P\left(\bigcap_{i=1}^{d} A_i \cap {B_i}^c\right) \\
=& P\left(\left(\bigcap_{i=1}^{d} A_i\right) \cap \left(\bigcap_{i=1}^{d} {B_i}^c\right)\right) \\
=& P\left(\left(\bigcap_{i=1}^{d} A_i\right) \cap \left(\bigcup_{i=1}^{d} B_i\right)^c\right) \\
=& P\left(\left(\bigcap_{i=1}^{d} A_i\right) \setminus \left(\bigcup_{i=1}^{d} B_i\right)\right) \\
=& P\left(\left(\bigcap_{i=1}^{d} A_i\right)\right) - P\left(\left(\bigcap_{i=1}^{d} A_i\right) \cap \left(\bigcup_{i=1}^{d} B_i\right)\right) \\
=& P\left(\left(\bigcap_{i=1}^{d} A_i\right)\right) - P\left(\left(\bigcup_{i=1}^{d} A_i \cap B_i\right)\right) \\
=& P\left(\left(\bigcap_{i=1}^{d} A_i\right)\right) + \sum_{J\subseteq\mathbb{N}_{\le d}\wedge J\ne\emptyset} (-1)^{|J|} P\Bigg(\Bigg(\bigcap_{j\in\mathbb{N}_{\le d}\setminus J} \\
=& \sum_{J\subseteq\mathbb{N}_{\le d}} (-1)^{|J|} P\left(\left(\bigcap_{j\in\mathbb{N}_{\le d}\setminus J} A_j\right) \cap \left(\bigcap_{j\in J} B_j\right)\right)
\end{aligned}$$

$$\sum_{(\varepsilon_1,\ldots,\varepsilon_d)\in\{0,1\}^d} (-1)^{d-\sum_{i=1}^d \varepsilon_i} F_X(a_1+\varepsilon_1(b_1-a_1),\ldots,a_d+\varepsilon_d(b_d-a_d)).$$

□

ix Rectangle increasing property of joint CDF

For any rectangle $R = \prod_{i=1}^{d}(a_i, b_i]$,

$$\Delta F_X(R) \geq 0.$$

x Converse property of CDF

If a function $F : \mathbb{R}^d \to [0,1]$ is such that for any $(a_1, \dots, a_d)$ and $(b_1, \dots, b_d)$ with $a_i \leq b_i$ for all $i \in \mathbb{Z} \cap [1, d]$,

$$F(a_1, \dots, a_d) \leq F(b_1, \dots, b_d),$$

for any $i \in \mathbb{Z} \cap [1, d]$,

$$\lim_{x_i \to -\infty} F(x_1, \dots, x_d) = 0,$$

$$\lim_{(x_1,\dots,x_d)\to(\infty,\dots,\infty)} F(x_1, \dots, x_d) = 1,$$

$$\lim_{x_1 \to {a_1}^+,\dots,x_d \to {a_d}^+} F(x_1, \dots, x_d) = F(a_1, \dots, a_d),$$

and, for any rectangle $R = \prod_{i=1}^{d}(a_i, b_i]$,

$$\Delta F(R) \geq 0,$$

then there exists a real-valued random vector X such that $F_X = F$.

xi Marginal CDF

The marginal CDF of the sub-vector $Y = (X_{i1}, \dots, X_{ik})$ is

$$\begin{aligned} F_Y(x_{i1}, \dots, x_{ik}) =& P(X_{i1} \leq x_{i1}, \dots, X_{ik} \leq x_{ik}) \\ =& \lim_{\forall j \in \mathbb{Z}\cap[1,d]\setminus\{i_l\,:\, l\in[1,k]\}\,:\, x_j \to \infty} F_X(x_1, \dots, x_d) \end{aligned}$$

XI Discrete random vector

i Discrete random vector

A random vector of discrete random variables.

ii (Joint, multivariable, or multivariate) probability mass function (PMF)

$$\begin{aligned} p_X(x_1, \dots, x_d) &= P(X_1 = x_1, \dots, X_d = x_d) \\ &= P(X_1 = x_1) \cdot P(X_2 = x_2 \mid X_1 = x_1) \cdot P(X_3 = x_3 \mid X_1 = x_1, X_2 = x_2) \cdot \\ &\dots \cdot P(X_d = x_d \mid X_1 = x_1, X_2 = x_2 \dots X_{d-1} = x_{d-1}) \end{aligned}$$

iii Marginal PMF

The marginal PMF (or sometimes simply marginal) of the sub-vector $Y = (X_{i1}, \dots, X_{ik})$ is

$$\begin{aligned} p_Y(x_{i1}, \dots, x_{ik}) =& P(X_{i1} = x_{i1}, \dots, X_{ik} = x_{ik}) \\ =& \sum_{\forall j \in \mathbb{Z}\cap[1,d]\setminus\{i_l\,:\, l\in[1,k]\}\,:\, x_j} p_X(x_1, \dots, x_d) \end{aligned}$$

iv Support

$$\mathrm{Supp}(X) = \prod_{i=1}^{d} \mathrm{Supp}(X_i).$$

v Properties

Let target sample space be Ω and target event space be $\mathcal{F}$.

$$p_X(\mathbf{a}) \geq 0, \quad \forall \mathbf{a} \in \Omega.$$
$$\sum_{\mathbf{a}\in\Omega} p_X(\mathbf{a}) = 1.$$
$$P(X \in A) = \sum_{\mathbf{a}\in A} p_X(\mathbf{a}), \quad \forall A \in \mathcal{F}_X.$$
$$\nabla F_X = \mathbf{0} \quad \text{a.e.}.$$

XII Absolutely Continuous Random Vector

i Absolutely continuous random vector

A random vector of absolutely continuous random variables.

ii (Joint, multivariable, or multivariate) probability density function (PDF)

$$f_X(x_1, \dots, x_d) = \frac{\partial^n F_X(x_1, \dots, x_d)}{\partial x_1 \dots \partial x_d}.$$

Equality of joint PDF means equality a.e.

iii Marginal PDF

For the sub-vector $Y = (X_{i1}, \dots, X_{ik})$, suppose marginal CDF of it is $F_Y(x_{i1}, \dots, x_{ik})$, the marginal PDF (or sometimes simply marginal) is

$$\begin{aligned} f_Y(X_{i1}, \dots, X_{ik})(x_{i1}, \dots, x_{ik}) =& \frac{\partial^n F_Y(x_{i1}, \dots, x_{ik})}{\partial x_{i1} \dots \partial x_{ik}} \\ =& \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_X(x_1, \dots, x_d) \prod_{j\in\mathbb{Z}\cap[1,d]\setminus\{i_l\,:\; l\in[1,k]} \mathrm{d}x_j \end{aligned}$$

iv Support

$$\mathrm{Supp}(X) = \prod_{i=1}^{d} \mathrm{Supp}(X_i).$$

v Properties

$$f_X(\mathbf{x}) \geq 0.$$
$$\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_X(x_1, \dots, x_d)\, \mathrm{d}x_1 \dots \mathrm{d}x_d = 1.$$
$$P(X \in B) = \int_B f_X(x_1, \dots, x_d)\, \mathrm{d}x_1 \dots \mathrm{d}x_d\,, \quad \forall B \in \mathcal{B}(\mathbb{R}).$$

XIII Derived Random Vector

i Derived random variable

A real-valued random variable Y defined as $Y = g(X_1, \dots, X_n)$ for some real-valued random variables $X_1, \dots, X_n$ and measurable function $g : \mathbb{R}^n \to \mathbb{R}$ is called a derived random variable.

ii Derived random vector

A real-valued random vector $Y = (Y_1, \dots, Y_m)$ defined as $Y = g(X)$ for some real-valued random vector $X = (X_1, \dots, X_n)$ and measurable function $g : \mathbb{R}^n \to \mathbb{R}^m$ is called a derived random vector.

iii Maximum and minimum of random variables

Consider a real-valued random vector $X = (X_1, \dots, X_d)$, and define $Y = \max\left(X_1, \dots, X_d\right)$ and $Z = \min\left(X_1, \dots, X_d\right)$.

$$F_Y(y) = P(X_1 \leq y \wedge \dots \wedge X_d \leq y) = F_X(y, \dots, y)$$

$$\begin{aligned} F_Z(z) =& P(X_1 \leq z \vee \dots \vee X_d \leq z) \\ =& 1 - P(X_1 > z, \dots, X_d > z) \\ =& 1 - \prod_{i=1}^{d} \left(1 - F_{X_i}(z)\right) \\ =& \sum_{J \subseteq \mathbb{N}_{\leq d} \wedge J \neq \emptyset} (-1)^{|J|+1} P\left(X_j \leq z \forall j \in J\right) \\ =& \sum_{J \subseteq \mathbb{N}_{\leq d} \wedge J \neq \emptyset} (-1)^{|J|+1} F_{(X_j)_{j \in J}}(z, \dots, z) \end{aligned}$$

iv CDF of derived random vector

$$F_Y(y_1, \dots, y_m) = P(Y_1 \leq y_1, \dots, Y_m \leq y_m)$$

v PMF of derived random vector

$$p_Y(y_1, \dots, y_m) = P(Y_1 = y_1, \dots, Y_m = y_m)$$

vi PDF of derived random vector

$$f_Y(y_1, \dots, y_m) = \frac{\partial^m}{\partial y_1 \dots \partial y_m} F_Y(y_1, \dots, y_m)$$

vii Special case of injective function

Let $X = (X_1, \dots, X_n)$ be an absolutely continuous random vector, $g : \mathbb{R}^n \to \mathbb{R}^n$ be injective and differentiable, and C^1 a.e., $\left|\det(J(g))\right|$ be nonzero, g^{-1} be differentiable, and C^1 a.e., and $Y = g(X)$. Then

$$f_Y(y) = f_X(g^{-1}(y)) \left|\det\left((J(g^{-1}))(y)\right)\right| .$$

viii Affine transformation

Let $X = (X_1, \dots, X_n)$ be an absolutely continuous random vector, A be an invertible $n \times n$ matrix, and b be an m-dimensional vector, and $Y = AX + b$. Then

$$f_Y(y) = \frac{1}{\det(A)} f_X(A^{-1}(y - b)).$$

XIV Independent random variables

i Definition

real-valued random variables $X_1, \dots, X_n$ are (mutually) independent if for all Borel sets $A_1, \dots, A_n$,

$$P(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^{n} P(X_i \in A_i).$$

ii π-system

real-valued random variables $X_1, \dots, X_n$ are (mutually) independent if there exists π-system I on the real numbers that generates the Borel algebra such that for all sets $A_1, \dots, A_n \in I$,

$$P(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^{n} P(X_i \in A_i).$$

Proof. By measures agree on generating π-system agree theorem. □

A particularly useful π-system on the real numbers that generates the Borel algebra is the set C of all $(-\infty, x]$ with $x \in \mathbb{R}$. And when $\mathbb{R}$ is restricted to a countable subspace $U \subseteq \mathbb{R}$ with subspace topology, C is restricted to the power set of U and generates the Borel algebra of U, which is also the power set of U.

iii Subset independence

If real-valued random variables $X_1, \dots, X_n$ are independent, then for any permutation σ on $\mathbb{Z} \cap [1, n]$, for any positive integer $m \leq n$, $X_{\sigma(1)}, \dots, X_{\sigma(m)}$ are independent.

iv CDF factorization

$X_1, \dots, X_d$ are independent iff

$$F_X(x_1, \dots, x_d) = \prod_{i=1}^{d} F_{X_i}(x_i).$$

Proof. Only if: Trivial.

If: By π-system of $(-\infty, x]$ with $x \in \mathbb{R}$. □

v PMF factorization

$X_1, \dots, X_d$ are independent iff

$$p_X(x_1, \dots, x_d) = \prod_{i=1}^{d} p_{X_i}(x_i).$$

Proof. Only if: Trivial.

If: By subspace π-system. □

vi PDF factorization

$X_1, \dots, X_d$ are independent iff

$$f_X(x_1, \dots, x_d) = \prod_{i=1}^{d} f_{X_i}(x_i).$$

Proof. Only if: Trivial.

If: By CDF factorization and absolute continuity. □

vii Independent and identically distributed (i.i.d.) random variables

$X_1, \dots, X_d$ are said to be independent and identically distributed (i.i.d.) if they are independent and the marginal CDFs F_{X_i} for all $i \in \mathbb{Z} \cap [1, d]$ are identical.

viii Functions of independent random variables are independent

Let $X_1, \dots, X_n$ be independent real-valued random variables and $g_1, \dots, g_n : \mathbb{R} \to \mathbb{R}$ be measurable functions. Then $g_1(X_1), \dots, g_n(X_n)$ are independent.

Proof. For Borel sets $A_1, \dots, A_n$,

$$\begin{aligned} P(g_1(X_1) \in A_1, \dots, g_n(X_n) \in A_n) &= P(X_1 \in {g_1}^{-1}(A_1), \dots, X_n \in {g_n}^{-1}(A_n)) \\ &= \prod_{i=1}^{n} P(X_i \in {g_i}^{-1}(A_i)) \\ &= \prod_{i=1}^{n} P(g_i(X_i) \in A_i) \end{aligned}$$

□

XV Product of random variable

i Definition

For two real-valued random variables X, Y with target sample space Ω, the product of random variable XY is defined with

$$(XY)(\omega) = X(\omega)Y(\omega), \quad \forall \omega \in \Omega.$$

ii For discrete random variable

For two discrete random variables X, Y with target sample space Ω, for all $z \in \Omega$,

$$p_{XY}(z) = \sum_{x, z-x \in \Omega} P(X = x, Y = z - x).$$

iii For absolutely continuous random variable

For two absolutely continuous random variables X, Y,

$$f_{XY}(z) = \int_{-\infty}^{0} f_{X,Y}\left(x, \frac{z}{x}\right) \frac{1}{|x|} \,\mathrm{d}x + \int_{0}^{\infty} f_{X,Y}\left(x, \frac{z}{x}\right) \frac{1}{|x|} \,\mathrm{d}x\,.$$

XVI (Mathematical) Expected Value, Expectation (期望值), or Mean (平均值)

i Definition

For a real-valued random variable X on $(\Omega_s, \mathcal{F}_s, P_s)$ with induced probability space $(\Omega_X, \mathcal{F}_X, P_X)$, the expected value or expectation $E(X)$, $\mathbb{E}(X)$, $E[X]$, or $\mathbb{E}[X]$, or mean μ_X, of it is defined as the Lebesgue integral

$$E(X) = \int_{\Omega_s} X(\omega_s) \,\mathrm{d}P_s(\omega_s)\,,$$

or equivalently,

$$E(X) = \int_{\Omega_X} x \,\mathrm{d}P_X(x)\,,$$

or in Stieltjes integral

$$E(X) = \int_{x=-\infty}^{\infty} x \,\mathrm{d}F_X(x)\,.$$

ii For discrete random variable

For a discrete random variable X with target sample space Ω_X and probability mass function $p_X(x)$, the expected value or expectation $E(X)$, $\mathbb{E}(X)$, $E[X]$, or $\mathbb{E}[X]$, or mean μ_X, of it is

$$E(X) = \sum_{x \in \Omega_X} x p_X(x)$$

if Ω_X is a finite set or Ω_X is a countably infinite set and

$$\sum_{x \in \Omega_X} |x|\, p_X(x)$$

is convergent, and doesn't exist if Ω_X is a countably infinite set and

$$\sum_{x \in \Omega_X} |x|\, p_X(x)$$

is divergent.

iii For absolutely continuous random variable

For an absolutely continuous random variable X with probability density function $f_X(x)$, the expected value or expectation $E(X)$, $\mathbb{E}(X)$, $E[X]$, or $\mathbb{E}[X]$, or mean μ_X, of it is

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) \, \mathrm{d}x$$

if it converges absolutely.

iv Derived random variable theorem - unary case

For a real ramdom variable and a unary function $g \in L^1(\Omega_X, P_X)$,

$$E(g(X)) = \int_{\Omega_X} g(x) \, \mathrm{d}P_X(x) \, ,$$

that is,

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x) \, \mathrm{d}x$$

for absolutely continuous random variable and

$$E(g(X)) = \sum_{x \in \mathrm{Supp}(X)} g(x) p_X(x)$$

for discrete random variable.

v Derived random variable theorem - general case

For a real-valued random vector $X = (X_1, \dots, X_d)$ with target sample spaces $\Omega_1, \dots, \Omega_d$, pushforward probability measures $P_1, \dots, P_d$, and an n-ary function $g \in L^1(\prod_{i=1}^{d} \Omega_i, P)$, where P is the product measure in the product space $\prod_{i=1}^{d} \Omega_i$,

$$E(g(X)) = \int_{\prod_{i=1}^{d} \Omega_i} g(x_1, \dots, x_d) \, \mathrm{d}P(x_1, \dots, x_d) = \int_{\Omega_1} \dots \int_{\Omega_d} g(x_1, \dots, x_d) \, \mathrm{d}P_1(x_1) \dots \mathrm{d}P_d(x_d) \, ,$$

that is,

$$E(g(X)) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} g(x_1, \dots, x_d) f_X(x_1, \dots, x_d) \, \mathrm{d}x_1 \dots \mathrm{d}x_n$$

for absolutely continuous random vector and

$$E(g(X)) = \sum_{(x_1, \dots, x_d) \in \mathrm{Supp}(X)} g(x_1, \dots, x_d) p_X(x_1, \dots, x_d)$$

for discrete random vector.

vi Expected value from CDF

For a non-negative real-valued random variable X,

$$E(X) = \int_0^\infty P(X \geq x)\,\mathrm{d}x = \int_0^\infty 1 - F_X(x)\,\mathrm{d}x\,.$$

Proof.

$$\begin{aligned} E(X) &= \int_{\Omega_X} x\,\mathrm{d}P_X(x) \\ &= \int_{\Omega_X}\int_0^x 1\,\mathrm{d}t\,\mathrm{d}P_X(x) \\ &= \int_0^\infty \int_{\Omega_X} \mathbf{1}_{t\in[0,x)}(x,t)\,\mathrm{d}P_X(x)\,\mathrm{d}t \\ &= \int_0^\infty P(X \geq t)\,\mathrm{d}t \end{aligned}$$

□

For real-valued random variable X, let

$$X^+ = \begin{cases} X, & X > 0 \\ 0, & X \leq 0 \end{cases}$$

and

$$X^- = \begin{cases} -X, & X < 0 \\ 0, & X \geq 0 \end{cases}.$$

Then,

$$E(X^+) = \int_0^\infty P(X \geq x)\,\mathrm{d}x = \int_0^\infty 1 - F_X(x)\,\mathrm{d}x\,,$$

$$E(X^-) = \int_{-\infty}^0 P(X \leq x)\,\mathrm{d}x = \int_{-\infty}^0 F_X(x)\,\mathrm{d}x\,,$$

Proof.

$$\begin{aligned} E(X^-) &= \int_0^\infty P(X^- \geq u)\,\mathrm{d}u \\ &= \int_0^\infty P(X \leq -u)\,\mathrm{d}u\,, \quad x = -u \\ &= \int_{-\infty}^0 P(X \leq x)\,\mathrm{d}x \\ &= \int_{-\infty}^0 F_X(x)\,\mathrm{d}x \end{aligned}$$

□

and

$$E(X) = E(X^+) - E(X^-).$$

vii Product of random variables

For d real-valued random variables $X_1, X_2, \dots, X_d$,

$$E(\prod_{i=1}^{d} X_i) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{i=1}^{d} x_i f_{X_1, X_2, \dots, X_d}(x_1, x_2, \dots, x_d)\,\mathrm{d}x_1 \dots \mathrm{d}x_d\,.$$

viii Product of independent random variables

If $X_1, \dots, X_d$ are independent,

$$E(\prod_{i=1}^{d} X_i) = \prod_{i=1}^{d} E(X_i).$$

ix Affine Transformation

For a real-valued random variable or real-valued random vector X, $a \in \mathbb{R}_{\neq 0}$, and $b \in \mathbb{R}$,

$$E(aX + b) = aE(X) + b.$$

x Linearity

For indexed families of real-valued random variables or real-valued random vectors $(X_i)_{i \in I}$ and real numbers $(a_i)_{i \in I}$ with $|I| \in \mathbb{N} \cup \{\aleph_0\}$, where for $|I| = \aleph_0$ suppose RHS is convergent,

$$E(\sum_{i \in I} a_i X_i) = \sum_{i \in I} a_i E(X_i).$$

xi Conditional expectation

$$E(X|A) = \frac{E(X \cdot \mathbf{1}_A)}{P(A)}$$

xii Law of total expectation

Let $(A_i)_{i=1}^{n} \subseteq \mathcal{F}$ be a partition. Then

$$E(X) = \sum_{i=1}^{n} E(X|A_i)P(A_i)$$

XVII Variance and Covariance

i Variance (變異數/方差)

For a real-valued random variable X, variance $\mathrm{Var}(X)$, $V(X)$, or ${\sigma_X}^2$ of it is defined as

$$\mathrm{Var}(X) = E((X - E(X))^2) = E(X^2) - 2E(X E(X)) + E(X)^2 = E(X^2) - 2E(X)^2 + E(X)^2 = E(X^2) - E(X)^2.$$

ii Standard Deviation (標準差)

For a real-valued random variable X, standard deviation $\mathrm{SD}(X)$ or σ_X of it is defined as

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

iii Covariance (共變異數/共方差) and covariance matrix

For two real-valued random variables X, Y, their covariance $\Sigma(X, Y)$ or $\mathrm{Cov}(X, Y)$ is defined as

$$\mathrm{Cov}(X, Y) = E\left((X - E(X))(Y - E(Y))\right) = E(XY) - E(X)E(Y).$$

Thus,

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X).$$

If X, Y are independent,

$$\mathrm{Cov}(X, Y) = 0.$$

For an n-dimensional vector of real-valued random variables $X = (X_1, X_2, \dots)$ with $n \in \mathbb{N} \cup \{\aleph_0\}$, the covariance matrix $\Sigma(X)$, $\mathrm{Cov}(X)$, or $\mathrm{CovMat}(X)$ is defined as

$$\mathrm{CovMat}(X) = E\left((\mathbf{X} - E(\mathbf{X}))(\mathbf{X} - E(\mathbf{X}))^\top\right),$$

that is,

$$\mathrm{CovMat}(X)_{ij} = \mathrm{Cov}\left(X_i, X_j\right) = E\left((X_i - E(X_i))(X_j - E(X_j))\right) = E(X_iX_j) - E(X_i)E(X_j).$$

Thus, for a real-valued random variable X,

$$\mathrm{CovMat}(X, \dots, X)_{ij} = \mathrm{Var}(X).$$

iv Zero variance

Variance of a real-valued random variable X is zero iff X is almost surely a constant, that is, there exists a constant c such that $P(X = c) = 1$.

If X has zero variance, then for any real-valued random variable Y,

$$\mathrm{Cov}(X, Y) = 0.$$

v Mean square error (MSE) (均方（誤）差) or Expected square error

For a real-valued random variable X and a prediction, predictor, or (blind) estimate $\hat{x}$, the mean square error $\mathrm{MSE}(\hat{x})$ or expected square error of the prediction is defined as

$$\mathrm{MSE}(\hat{x}) = E\left((X - \hat{x})^2\right).$$

vi Minimum mean square error

The minimum mean square error estimate of a real-valued random variable X is $E(X)$, and the MSE of it is $\mathrm{Var}(X)$.

vii Pearson correlation coefficient (PCC), Pearson's r, Pearson product-moment correlation coefficient (PPMCC), or correlation coefficient

For two real-valued random variables X, Y, the Pearson correlation coefficient (PCC), Pearson's r, Pearson product-moment correlation coefficient (PPMCC), or correlation coefficient of them, $\rho_{X,Y}$, is defined as

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X)\,\mathrm{SD}(Y)} = \mathrm{Cov}\left(\frac{X - E(X)}{\mathrm{SD}(X)}, \frac{Y - E(Y)}{\mathrm{SD}(Y)}\right).$$

Thus,

$$\rho_{X,X} = 1$$

If X, Y are independent,

$$\rho_{X,Y} = 0.$$

viii Correlation

For two real-valued random variables X, Y, the correlation $r_{X,Y}$ is defined as

$$r_{X,Y} = E(XY).$$

Thus,

$$r_{X,X} = E(X^2).$$

If X, Y are independent,

$$r_{X,Y} = E(XY) = E(X)E(Y).$$

ix Sum of random variables

For n-dimensional real-valued random vector $X = (X_1, X_2, \dots, X_n)$ and real vector $a = (a_1, a_2, \dots, a_n)$ with $n \in \mathbb{N}$,

$$\mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = a^\top \,\mathrm{CovMat}(X) a = \sum_{i=1}^{n} a_i^2 \,\mathrm{Var}\big(X_i\big) + 2a_i a_j \sum_{i=1}^{n} \sum_{j=i+1}^{n} \mathrm{Cov}\big(X_i, X_j\big).$$

For real-valued random variables X, Y and real numbers a, b,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y).$$

For real ranom variables $X_1, \dots, X_c, Y_1, \dots, Y_d$ and real numbers $a_1, \dots,_c, b_1, \dots, b_d$,

$$\mathrm{Cov}\left(\sum_{i=1}^{c} a_i X_i, \sum_{i=1}^{d} b_i Y_i\right) = \sum_{i=1}^{c} \sum_{j=1}^{d} a_i b_j \,\mathrm{Cov}\big(X_i, Y_j\big)$$

(bilinearity).

x Affine Transformation

For a real-valued random variable X, $a \in \mathbb{R}$, and $b \in \mathbb{R}$, let $W = aX + b$,

$$\mathrm{Var}(W) = a^2 \,\mathrm{Var}(X).$$

$$\mathrm{SD}(W) = |a| \,\mathrm{SD}(X).$$

$$\rho_{W,X} = \begin{cases} \mathrm{sgn}(a), & a \neq 0 \\ 0, & a = 0 \end{cases}.$$

For two real-valued random variables X, Y, $a, c \in \mathbb{R}$, and $b, d \in \mathbb{R}$, let $U = aX + b$ and $V = cY + b$,

$$\mathrm{Cov}(U, V) = ac \,\mathrm{Cov}(X, Y).$$

$$\rho_{U,V} = \begin{cases} \mathrm{sgn}(a)\,\mathrm{sgn}(c)\rho_{X,Y}, & ac \neq 0 \\ 0, & ac = 0 \end{cases}.$$

xi Orthogonal random variables

Two real-valued random variables X, Y are orthogonal if $r_{X,Y} = 0$.

xii Uncorrelated random variables

Two real-valued random variables X, Y are uncorrelated if $\text{Cov}(X, Y) = 0$.

xiii Converse of independent implies uncorrelated is not true

Two real-valued random variables X, Y such that $\text{Cov}(X, Y) = 0$ are not necessarily independent.

Proof. A counterexample is $X \sim N(0, 1)$ and $Y = X^2$. □

xiv Uncorrelated random variables properties

For n-dimensional real-valued random vector $X = (X_1, X_2, \ldots, X_d)$ of pairwise uncorrelated random variables, $\text{CovMat}(X)$ is a diagonal matrix where $\text{CovMat}(X)_{ii} = \text{Var}\left(X_i\right)$, and

$$\text{Var}\left(\sum_{i=1}^{d} X_i\right) = \sum_{i=1}^{d} \text{Var}\left(X_i\right).$$

xv Correlation coefficient range

$$-1 \leq \rho_{X,Y} \leq 1$$

Proof. For any fixed constant a, let $W = X - aY$. Then,

$$\begin{aligned}
\text{Var}(W) =& E((X - aY)^2) - (E(X - aY))^2 \\
=& E(X^2 - 2aXY + a^2Y^2) - \left(E(X)^2 - 2aE(X)E(Y) + a^2E(Y)\right)^2 \\
=& \text{Var}(X) - 2a\,\text{Cov}(XY) + a^2\,\text{Var}(Y)
\end{aligned}$$

Since $\text{Var}(W) \geq 0$, we have

$$2a\,\text{Cov}(X, Y) \leq \text{Var}(X) + a^2\,\text{Var}(Y)$$

Choosing

$$a = \frac{\sigma_X}{\sigma_Y}$$

yields

$$\text{Cov}(X, Y) \leq \sigma_X \sigma_Y.$$

Choosing

$$a = -\frac{\sigma_X}{\sigma_Y}$$

yields

$$\text{Cov}(X, Y) \geq -\sigma_X \sigma_Y.$$

□

When $\rho_{X,Y} > 0$, X and Y are called positively correlated. When $\rho_{X,Y} < 0$, X and Y are called negatively correlated. When $\rho_{X,Y} = 0$, X and Y are uncorrelated.

xvi Best linear predictor

The best linear predictor $\hat{Y}$ of a real-valued random variable Y based on another real-valued random variable X in terms of mean squared error $E((Y-\hat{Y})^2)$ is given by $\hat{Y} = aX + b$, where

$$a = \begin{cases} \dfrac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)} = \rho_{X,Y}\dfrac{\mathrm{SD}(Y)}{\mathrm{SD}(X)}, & \mathrm{Var}(X) \neq 0, \\ \text{arbitrary}, & \mathrm{Var}(X) = 0, \end{cases}$$

$$b = E(Y) - aE(X),$$

and the mean squared error $E((Y-aX-b)^2)$ then is

$$E((Y-aX-b)^2) = \left(1 - {\rho_{X,Y}}^2\right)^2 \mathrm{Var}(Y).$$

When $\rho_{X,Y} = 1$, $E((Y-aX-b)^2) = 0$ and $Y = \hat{Y}$.

Proof. Let

$$X' = X - E(X)$$
$$Y' = Y - E(Y)$$
$$Y - aX - b = (Y' - aX') + (E(Y) - aE(X) - b)$$
$$E((Y-aX-b)^2) = E((Y'-aX')^2) + 2(E(Y) - aE(X) - b)E(Y' - aX') + (E(Y) - aE(X) - b)^2$$
$$E(X') = E(Y') = E(Y' - aX') = 0$$
$$E((Y-aX-b)^2) = E((Y'-aX')^2) + (E(Y) - aE(X) - b)^2$$
$$b = E(Y) - aE(X)$$
$$\begin{aligned} E((Y' - aX' - (E(Y) - aE(X)))^2) =& E((Y'-aX')^2) \\ =& E(Y'^2) - 2aE(X'Y') + a^2E(X'^2) \\ =& \mathrm{Var}(Y) - 2a\,\mathrm{Cov}(X,Y) + a^2\,\mathrm{Var}(X) \end{aligned}$$
$$\frac{\mathrm{d}\left(\mathrm{Var}(Y) - 2a\,\mathrm{Cov}(X,Y) + a^2\,\mathrm{Var}(X)\right)}{\mathrm{d}a} = -2\,\mathrm{Cov}(X,Y) + 2a\,\mathrm{Var}(X) = 0$$
$$a = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}$$
$$\begin{aligned} \mathrm{Var}(Y) - 2a\,\mathrm{Cov}(X,Y) + a^2\,\mathrm{Var}(X) =& \mathrm{Var}(Y) - \frac{\mathrm{Cov}(X,Y)^2}{\mathrm{Var}(X)} \\ =& \mathrm{Var}(Y)\left(1 - \frac{\mathrm{Cov}(X,Y)^2}{\mathrm{Var}(X)\,\mathrm{Var}(Y)}\right) \\ =& \left(1 - {\rho_{X,Y}}^2\right)^2 \mathrm{Var}(Y) \end{aligned}$$

□

XVIII Moment, Moment generating function (MGF), and Characteristic function (CF)

i Moment

The nth moment of X is defined as $E(X^n)$ for $n \in \mathbb{N}$. The zero-th moment $E(X^0)$ is defined as 1.

ii Central moment

The nth central moment of X is defined as $E((X - E(X))^n)$. The zero-th central moment $E((X - E(X))^0)$ is defined as 1.

iii Moments determine finite-range random variable theorem

A random variable X with finite range is uniquely determined by the zero-th to $(n-1)$th moments $\mu_0 = 1, \mu_1, \dots, \mu_{n-1}$.

Proof. Let the range of X be $\{x_1, x_2, \dots, x_n\}$, moments be $\{\mu_0 = 1, \mu_1, \dots, \mu_{n-1}\}$, and $P(X = x_i) = p_i$. Then

$$\begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \\ {x_1}^2 & {x_2}^2 & \dots & {x_n}^2 \\ \vdots & \vdots & \ddots & \vdots \\ {x_1}^{n-1} & {x_2}^{n-1} & \dots & {x_n}^{n-1} \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{bmatrix} = \begin{bmatrix} \mu_0 \\ \mu_1 \\ \vdots \\ \mu_{n-1} \end{bmatrix}$$

□

iv Moment generating function (MGF)

The moment generating function (MGF) of a random variable X, denoted as $M_X(t)$, is defined as

$$M_X(t) = E\left(e^{tX}\right), \quad t \in \mathbb{R}.$$

v MGF as Laplace transform

In Stieltjes integral,

$$M_X(t) = \int_{x=-\infty}^{\infty} e^{tx} \, \mathrm{d}F_x(x).$$

vi MGF determines CDF

If the MGF of X exists in a neighborhood of $t = 0$, it uniquely determines the distribution of X.

Proof. By Mellin's inverse formula. □

vii MGF generating moments

If the MGF of X exists in a neighborhood of $t = 0$, $E(X^n) = {M_X}^{(n)}(0)$.

Proof. Suppose the MGF of X exists in $(-h, \varepsilon)$ for some $\varepsilon > 0$ and $J = \left(-\frac{h}{2}, \frac{h}{2}\right)$.

For any $\delta > 0$ and $n \in \mathbb{N}_0$, there exists $C_n > 0$ such that

$$|X|^n \leq C_n e^{\delta |X|}$$

Take some δ such that $0 < \delta < \frac{h}{2}$,

$$\left| \frac{\mathrm{d}^n}{\mathrm{d}t^n} e^{tX} \right| = |X|^n e^{tX} \leq C_n e^{(\delta + t)|X|} \leq C_n e^{\frac{h}{2}|X|} \leq C_n \left(e^{\frac{h}{2}X} + e^{-\frac{h}{2}X} \right)$$

Thus, $\frac{\mathrm{d}^n}{\mathrm{d}t^n}e^{tX}$ is uniformly bounded in J.

By Leibniz integral rule, since the integrands are uniformly bounded and thus the integrals are finite, for $t \in J$,

$$\frac{\mathrm{d}}{\mathrm{d}t}E\left(e^{tX}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}} e^{tX}\,\mathrm{d}P(X) = \int_{\mathbb{R}} \frac{\mathrm{d}}{\mathrm{d}t}e^{tX}\,\mathrm{d}P(X) = E(\frac{\mathrm{d}}{\mathrm{d}t}e^{tX})$$

Apply Leibniz integral rule iteratively, we obtain

$$\frac{\mathrm{d}^n}{\mathrm{d}t^n}E\left(e^{tX}\right) = E\left(\frac{\mathrm{d}^n}{\mathrm{d}t^n}e^{tX}\right)$$

Thus,

$$M_X{}^{(n)}(t) = E(X^n e^{tX}) = \sum_{k=n}^{\infty}\frac{t^{k-n}}{(k-n)!}E(X^k)$$

$$M_X{}^{(n)}(0) = E(X^k)$$

□

viii Characteristic function (CF)

The moment generating function (MGF) of a random variable X, denoted as $\varphi_X(t)$, is defined as

$$\varphi_X(t) = E\left(e^{itX}\right), \quad t \in \mathbb{R}.$$

ix Affine Transformation

For a real-valued random variable X, $a \in \mathbb{R}_{\neq 0}$, and $b \in \mathbb{R}$,

$$M_{aX+b}(t) = e^{bt}M_X(at).$$

x Logarithmic Linearity

For indexed families of real-valued random variables $(X_i)_{i\in I}$ and real numbers $(a_i)_{i\in I}$ with $|I| \in \mathbb{N} \cup \{\aleph_0\}$, where for $|I| = \aleph_0$ suppose RHS is convergent,

$$M_{\sum_{i\in I} a_i X_i}(t) = \prod_{i\in I} M_{X_i}\left(a_i t\right).$$

xi Moments of poduct of independent random variables

If $X_1, \ldots, X_d$ are independent,

$$E(\prod_{i=1}^{d} X_i{}^n) = \prod_{i=1}^{d} E(X_i{}^n)$$

xii MGF of sum of independent random variables

If $X_1, \ldots, X_d$ are independent,

$$M_{\sum_{i=1}^{d} X_i}(t) = \prod_{i=1}^{d} M_{X_i}(t)$$

XIX Mode (眾數)

i For discrete random variable

For a discrete random variable X with PMF $p_X(x)$, Mode(X) is defined as

$$\text{Mode}(X) = \arg\max_x p_X(x),$$

which is not necessarily unique.

ii For absolutely continuous random variable

For an absolutely continuous random variable X with PDF $f_X(x)$, Mode(X) is defined as

$$\text{Mode}(X) = \arg\max_x f_X(x),$$

which is not necessarily unique.

iii Affine Transformation

For a real-valued random variable X, $a \in \mathbb{R}_{\neq 0}$, and $b \in \mathbb{R}$,

$$\text{Mode}(aX + b) = a\,\text{Mode}(X) + b.$$

XX Median (中位數)

i Definition

For a real variable X, the median Median(X) of it is define by

$$P(X \le \text{Median}(X)) \ge \frac{1}{2} \wedge P(X \ge \text{Median}(X)) \ge \frac{1}{2},$$

which is not necessarily unique.

When we denote Median(X) to be an interval I, it means that for all $x \in I$,

$$P(X \le x) \ge \frac{1}{2} \wedge P(X \ge x) \ge \frac{1}{2}.$$

ii Affine Transformation

For a real-valued random variable X, $a \in \mathbb{R}_{\neq 0}$, and $b \in \mathbb{R}$,

$$\text{Median}(aX + b) = a\,\text{Median}(X) + b.$$

XXI Entropy (熵) for discrete random variables

i Definition

For a discrete random variable X with target sample space Ω and PMF $p_X(x)$, the entropy $H(X)$ of base b is defined as

$$H(X) = E(-\log_b(p_X(x))) = -\sum_{x \in \Omega \wedge p_X(x) > 0} p_X(x) \log_b(p_X(x)),$$

where base $b = 2$ gives the unit of bits or shannons, base $b = e$ gives natural units or nat, and base $b = 10$ gives the unit of dits, bans, or hartleys. When not specified, $b = e$ is used.

ii Nonnegativity

$$H(X) \geq 0.$$

Proof. If $p_X(x) = 1$ for some x, $H(X) = 0$; otherwise, for all $x \in \Omega$ such that $p_X(x) > 0$, $p_X(x) \in (0,1)$, $\log_b\big(p_X(x)\big) \in (-\infty, 0)$, and $p_X(x)\log\big(p_X(x)\big) < 0$. □

iii Maximum entropy

If $|\Omega| = n \in \mathbb{N}$,

$$H(X) \leq \log_b n,$$

and $H(X) = \log_b n$ occurs when $p_X(x) = \frac{1}{n}$ for all $x \in \Omega$.

Proof.

$$H(X) = -\sum_{x \in \Omega \wedge p_X(x) > 0} p_X(x)\log_b\big(p_X(x)\big)$$

For each $x_i \in \Omega$, let $p_i = p_X(x_i)$,

$$H(X) = -\sum_{i=1}^{n} p_i \log_b\big(p_i\big)$$

$$\frac{\partial H(X)}{\partial p_i} = -\log_b\big(p_i\big) - 1$$

$$\sum_{i=1}^{n} p_i = 1$$

$$\frac{\partial \sum_{i=1}^{n} p_i - 1}{\partial p_i} = 1$$

$$-\log_b\big(p_i\big) - 1 - \lambda = 0$$

$$p_i = b^{-1-\lambda} \vee 0$$

Suppose m out of n p_i are not 0,

$$H(X) = -mb^{-1-\lambda}(-1-\lambda)$$

Thus when $m = n$, $H(X)$ has maximum, and $p_i = \frac{1}{n}$, and

$$H(X) = -n\frac{1}{n}(-\log_b n) = \log_b n$$

□

iv Affine Transformation

For a discrete random variable X, $a \in \mathbb{R}_{\neq 0}$, and $b \in \mathbb{R}$,

$$H(aX + b) = \begin{cases} H(X), & a \neq 0 \\ 0, & a = 0 \end{cases}.$$

XXII (Differential or continuous) entropy for absolutely continuous random variables

i Definition

For an absolutely continuous random variable X with target sample space Ω and PDF $f_X(x)$, the (differential or continuous) entropy $h(X)$ or $H(X)$ of base b is defined as

$$h(X) = E(-\log_b(f_X(x))) = -\int_\Omega f(x)\log_b(f(x))\,\mathrm{d}x\,,$$

where base $b = 2$ gives the unit of bits or shannons, base $b = e$ gives natural units or nat, and base $b = 10$ gives the unit of dits, bans, or hartleys. When not specified, $b = e$ is used. Differential entropy can be positive.

ii Affine Transformation

For an absolutely continuous random variable X, $a \in \mathbb{R}_{\neq 0}$, and $b \in \mathbb{R}$,

$$h(Y) = h(X) + \log|a|.$$

XXIII Conditioning

i Conditioning on the discrete level

For two real-valued random variables X, Y, for any $A \in \mathcal{F}_X$, for any $y \in \Omega_y$ such that $P(Y = y) > 0$,

$$P_{X|Y}(A|y) = P(X \in A|Y = y) = \frac{P(X \in A, Y = y)}{P(Y = y)}.$$

ii Conditioning using limit

For two real-valued random variables X, Y, for any $A \in \mathcal{F}_X$, for any $y \in \Omega_y$, we define

$$P_{X|Y}(A \mid y) = \lim_{h\to 0} \frac{P(A, y-h \leq Y \leq Y+h)}{P(y-h \leq Y \leq y+h)}$$

if the denominator is nonzero and the limit exists.

For any y such that $P(Y = y) > 0$, $P_{X|Y}$ defined here is the same as in conditioning on the discrete level.

For any fixed $A \in \mathcal{F}_X$, define a measure on $\mathcal{F}_Y$ $\mu_A = P(X \in A, Y \in \cdot)$, if $\mu_A \ll P_Y$, then $P_{X|Y}(A \mid y)$ is defined for P_Y-almost every $y \in \Omega_y$, and the function

$$P_{X|Y}(A \mid \cdot) : \Omega_y \to [0,1];\ y \mapsto P_{X|Y}(A \mid y)$$

is a Radon–Nikodym derivative $\frac{\mathrm{d}\mu_A}{\mathrm{d}P_Y}$, and thus

$$P_{X|Y} : \mathcal{F}_X \times \Omega_y \to [0,1];\ A, y \mapsto P_{X|Y}(A \mid y)$$

is a regular conditional probability.

Proof. PLACEHOLDER □

For any fixed $y \in \Omega_y$, if

$$\lim_{h\to 0} \frac{\int_{|t-y|<h} \left|P_{X|Y}(A \mid t) - P_{X|Y}(A \mid y)\right| \mathrm{d}P_Y(t)}{P_Y(|t-y|<h)} = 0$$

for every $A \in \mathcal{F}$, we can define a conditional random variable of X given $Y = y$, denoted as $X|Y = y$, with pushforward probability

$$P_{X|Y}(\cdot \mid y) : \mathcal{F}_X \to [0, 1].$$

If for any fixed $A \in \mathcal{F}_X$, $\mu_A \ll P_Y$, then any subset N of Ω_y such that for all $y \in N$, there exists some $A \in \mathcal{F}$ such that

$$\lim_{h\to 0} \frac{\int_{|t-y|<h} \left|P_{X|Y}(A \mid t) - P_{X|Y}(A \mid y)\right| \mathrm{d}P_Y(t)}{P_Y(|t-y|<h)} > 0,$$

is a P_Y-null set.

Proof. PLACEHOLDER □

Unless otherwise specified, we discuss conditioning using limit below.

iii Conditional CDF

We define the conditional CDF of $X|Y = y$ as

$$F_{X|Y}(x|y) = \lim_{h\to 0} P(X \le x \mid y-h \le Y \le y+h) = \lim_{h\to 0} \frac{P(X \le x, y-h \le Y \le y+h)}{P(y-h \le Y \le y+h)}.$$

If $P(Y = y) > 0$,

$$F_{X|Y}(x|y) = P(X \le x, Y = y).$$

iv Discrete case

If $F_{X|Y}$ is such that there exists a finite or countable set $A \subseteq \mathbb{R} \times (0, 1]$ such that

$$\sum_{(a,p)\in A} p = 1$$

and

$$F_{X|Y}(x|y) = \sum_{(a,p)\in A} p\mathbf{1}_{[a,\infty)}(x),$$

we can define the conditional PMF of $X|Y = y$ as

$$p_{X|Y}(x|y) = F_{X|Y}(x|y) - \lim_{a\to x^-} F_{X|Y}(a|y),$$

equivalently,

$$p_{X|Y}(x|y) = \lim_{h\to 0} P(X = x \mid y-h \le Y \le y+h).$$

If $P(Y = y) > 0$,

$$p_{X|Y}(x|y) = P(X = x \mid Y = y).$$

The support $\mathrm{Supp}(X|Y)$ of $X|Y$ is defined as the support of $p_{X|Y}$.

v Absolutely continuous case

If $F_{X|Y}$ is absolutely continuous, we can define the conditional PDF of $X|Y = y$ as

$$f_{X|Y}(x|y) = \frac{\mathrm{d}}{\mathrm{d}x} F_{X|Y}(x|y).$$

The support $\mathrm{Supp}(X|Y)$ of $X|Y$ is defined as the support of $f_{X|Y}$.

vi Both discrete

If X and Y are both discrete, conditioning on discrete level is enough to define $P_{X|Y}$ for P_Y-almost every $y \in \Omega_y$, and

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}, \quad y \in \mathrm{Supp}(Y).$$

vii Both absolutely continuous

If X and Y are both absolutely continuous,

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \quad y \in \mathrm{Supp}(Y).$$

Proof.

$$\begin{aligned} F_{X|Y}(x|y) &= \lim_{h\to 0} \frac{P(X \leq x, y - h \leq Y \leq y + h)}{P(y - h \leq Y \leq y + h)} \\ &= \lim_{h\to 0} \frac{\int_{y-h}^{y+h} \int_{-\infty}^{x} f_{X,Y}(u, v)\,\mathrm{d}u\,\mathrm{d}v}{\int_{y-h}^{y+h} f_Y(v)\,\mathrm{d}v} \\ &= \frac{\int_{-\infty}^{x} f_{X,Y}(u, y)\,\mathrm{d}y}{f_Y(y)} \end{aligned}$$

By L'Hôpital's rule,

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{\mathrm{d}}{\mathrm{d}x} F_{X|Y}(x|y) \\ &= \frac{\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{-\infty}^{x} f_{X,Y}(u, y)\,\mathrm{d}y\right)}{f_Y(y)} \\ &= \frac{f_{X,Y}(x, y)}{f_Y(y)} \end{aligned}$$

□

viii Independent discrete random variables

If X and Y are independent and both discrete,

$$p_{X|Y}(x|y) = p_X(x)$$

for P_Y-almost every $y \in \Omega_y$.

Proof.

$$\begin{aligned}p_{X|Y}(x|y) =& \frac{p_{X,Y}(x)}{p_Y(x)}\\ =& \frac{p_X(x)p_Y(x)}{p_Y(x)}\\ =& p_X(x)\end{aligned}$$

□

ix Independent absolutely continuous random variables

If X and Y are independent and both absolutely continuous,

$$f_{X|Y}(x|y) = f_X(x)$$

for P_Y-almost every $y \in \Omega_y$.

Proof.

$$\begin{aligned}f_{X|Y}(x|y) =& \frac{f_{X,Y}(x)}{f_Y(x)}\\ =& \frac{f_X(x)f_Y(x)}{f_Y(x)}\\ =& f_X(x)\end{aligned}$$

□

x Conditional expectation given a random variable

For a derived random variable $g(X,Y)$ such that $g(X,Y)|Y = y \in L^1(\Omega_X, P_{X|Y}(x|y))$,

$$E(g(X,Y)|Y = y) = \int_{\Omega_X} g(x,y)\,\mathrm{d}P_{X|Y}(x|y)\,.$$

For absolutely continuous $X|Y = y$,

$$E(g(X,Y)|Y = y) = \int_{-\infty}^{\infty} g(x,y) f_{X|Y}(x|y)\,\mathrm{d}x\,.$$

For discrete $X|Y = y$,

$$E(g(X,Y)|Y = y) = \sum_{\mathrm{Supp}(X)} g(x,y) p_{X|Y}(x|y).$$

xi Law of total expectation given a random variable

For two real-valued random variables X, Y such that $P_{X|Y}$ is a regular conditional probability and a derived random variable $g(X,Y)$ such that $g(X,Y)|Y = y \in L^1(\Omega_X, P_{X|Y}(x|y))$,

$$E(E(g(X,Y)|Y)) = E(g(X,Y)).$$

Proof.

$$E(E(g(X,Y)|Y)) = \int_{\Omega_Y}\int_{\Omega_X} g(x,y)\,\mathrm{d}P_{X|Y}(x|y)\,\mathrm{d}P_Y(y) = \int_{\Omega_{X,Y}} g(X,Y)\,\mathrm{d}P_{X,Y}(x,y) = E(g(X,Y)) = E(g(X,Y)).$$

□

xii Conditional expectation

For two real-valued random variables X, Y, the conditional expectation $E(X|Y)$ is a random variable on $\mathcal{F}_Y$ defined as

$$(E(X|Y))(K) = E(X|Y \in K), \quad K \in \mathcal{F}_Y.$$

Suppose the set A is defined recursively by the conditions that

- X is in A,
- any real number is in A,
- any defined arithmetic operation of elements of A is in A, and
- any expectation of element of A is in A,

and the set B is defined as the expectation of any element of A. For any $h(X) \in B$, $h(X|Y)$ is a random variable on $\mathcal{F}_Y$ defined for any $K \in \mathcal{F}_Y$ by replacing all $E(\cdot)$ with $E(\cdot|Y \in K)$ in the definition of $h(X)$.

xiii Law of total probability

For two real-valued random variables X, Y (such that $P_{X|Y}$ is a regular conditional probability, otherwise RHS is not defined) and an event R of the random vector X, Y,

$$P((X, Y) \in R) = E(P(X \in R_Y \mid Y)),$$

where $R_Y = \{x : (x, Y) \in R\}$.

Proof.

$$\begin{aligned} E(P(X \in R_Y \mid Y)) &= \int_{\Omega_Y} \int_{R_y} \mathrm{d}P_{X|Y}(x \mid y) \, \mathrm{d}P_Y(y) \\ &= \int_R \mathrm{d}P_{X,Y}(x, y) \\ &= P((X, Y) \in R) \end{aligned}$$

□

xiv Conditional variance

For two real-valued random variables X, Y, the conditional variance $\mathrm{Var}(X|Y)$ is a random variable defined for any $K \in \mathcal{F}_Y$ as

$$\mathrm{Var}(X|Y)(K) = \mathrm{Var}(X|Y \in K) = E((X - E(X|Y \in K))^2 | Y \in K).$$

xv Law of total variance

For two real-valued random variables X, Y (such that $P_{X|Y}$ is a regular conditional probability, otherwise RHS is not defined),

$$\mathrm{Var}(X) = E(\mathrm{Var}(X|Y)) + \mathrm{Var}(E(X|Y)).$$

Proof.

$$
\begin{aligned}
\mathsf{Var}(X) =& E(X^2) - E(X)^2 \\
=& E(E(X^2|Y)) - E(E(X|Y))^2 \\
=& E(\mathsf{Var}(X|Y) + E(X|Y)^2) - E(E(X|Y))^2 \\
=& E(\mathsf{Var}(X|Y) + E(X|Y)^2) - E(E(X|Y))^2 \\
=& E(\mathsf{Var}(X|Y)) + E(E(X|Y)^2) - E(E(X|Y))^2 \\
=& E(\mathsf{Var}(X|Y)) + E(\mathsf{Var}(X|Y))
\end{aligned}
$$

□

XXIV Sum of independent random variables and Inequalities

i Sum of independent random variables

For two independent real-valued random variables X, Y with the same source sample space and event space, and with pushforward probability measure P_X and P_Y respectively, let the Schwartz distributions induced by P_X and P_Y be T_X and T_Y respectively. Then the Schwartz distribution T_{X+Y} induced by the pushforward probability measure P_{X+Y} of the sum random variable $X + Y$ is the distributional convolution of T_1 and T_2, that is,

$$T = T_1 * T_2.$$

Proof. By invertibility of Laplace transform and MGF of sum of independent random variables. □

ii Sum of independent discrete random variables

For two independent discrete random variables X, Y,

$$p_{X+Y}(z) = \sum_{x\in\mathbb{R}} p_X(x)p_Y(z - x).$$

Proof.

$$P(X + Y = z) = \sum_{x\in\mathbb{R}} P(X = x)P(Y = z - x).$$

□

iii Sum of independent absolutely continuous random variables

For two independent absolutely continuous random variables X, Y,

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x)\,\mathrm{d}x\,.$$

Proof.

$$F_{X+Y}(z) = \int_{-\infty}^{\infty}\int_{-\infty}^{z-x} f_X(x)f_Y(y)\,\mathrm{d}y\,\mathrm{d}x = \int_{-\infty}^{\infty} f_X(x)F_Y(z - x)\,\mathrm{d}x\,.$$

$$f_{X+Y}(z) = \frac{\mathrm{d}}{\mathrm{d}z}F_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x)\,\mathrm{d}x\,.$$

□

iv Random sum of random variables

Let N be a positive-integer-valued random variable. Let $X_1, X_2 \dots$ be a sequence of random variables. We are interested in the distribution of the random sum $S_N = \sum_{i=1}^{N} X_i$. Here, we will focus on the case where X_i's are i.i.d. and independent of N.

v MGF of random sum of i.i.d. random variables

Let N be a positive-integer-valued random variable. Let $X_1, X_2 \dots$ be a sequence of i.i.d. random variables independent from N and $X_i \sim X$ for all i. Let $S_N = \sum_{i=1}^{N} X_i$.

$$M_{S_N}(t) = M_N(\ln(M_X(t))).$$

Proof.

$$\begin{aligned} M_{S_N}(t) &= E(e^{tS_N}) \\ &= E(E(e^{tS_N} | N)) \\ &= E(M_X(t)^N) \\ &= E\left(e^{\ln(M_X(t))N}\right) \\ &= M_N(\ln(M_X(t))) \end{aligned}$$

□

XXV Families of Discrete Random Variables

i Discrete uniform random variable

- Notation: $X \sim \mathcal{U}\{a, b\}$, $X \sim U\{a, b\}$, $X \sim \text{Unif}\{a, b\}$, $X \sim \text{Uniform}\{a, b\}$, $X \sim \mathcal{U}\{a, a+1, \dots, b\}$, $X \sim U\{a, a+1, \dots, b\}$, $X \sim \text{Unif}\{a, a+1, \dots, b\}$, or $X \sim \text{Uniform}\{a, a+1, \dots, b\}$, read as X is discretely uniform distributed with parameters a and b.
- Parameters: $a, b \in \mathbb{Z} \wedge a \leq b$
- Support: $\mathbb{Z} \cap [a, b]$
- PMF:

$$p_X(k) = \frac{1}{b-a+1}, \quad k \in \mathbb{Z} \cap [a, b]$$

- CDF:

$$F_X(k) = \begin{cases} 0, & k < a \\ \dfrac{\lfloor k \rfloor - a + 1}{b - a + 1}, & a \leq k < b \\ 1, & k \geq b \end{cases}$$

- Mean:

$$E(X) = \frac{a+b}{2}$$

- Variance:

$$\text{Var}(X) = \frac{(b-a+1)^2 - 1}{12}$$

Proof.

$$\begin{aligned}
E(X^2) &= \sum_{k=a}^{b} \frac{k^2}{b-a+1} \\
&= \frac{1}{b-a+1} \sum_{k=0}^{b-a} (a+k)^2 \\
&= \frac{1}{b-a+1} \left((b-a+1)a^2 + 2a\frac{(b-a+1)(b-a)}{2} + \frac{(b-a+1)(b-a)(2b-2a+1)}{6} \right) \\
&= a^2 + a(b-a) + \frac{(b-a)(2b-2a+1)}{6} \\
&= ab + \frac{2b^2 - 2ab + b - 2ab + 2a^2 - a}{6} \\
&= \frac{2a^2 + 2b^2 + 2ab - a + b}{6}
\end{aligned}$$

$$\begin{aligned}
\text{Var}(X) &= E(X^2) - E(X)^2 \\
&= \frac{2a^2 + 2b^2 + 2ab - a + b}{6} - \frac{a^2 + b^2 + 2ab}{4} \\
&= \frac{4a^2 + 4b^2 + 4ab - 2a + 2b - 3a^2 - 3b^2 - 6ab}{12} \\
&= \frac{a^2 + b^2 - 2ab - 2a + 2b}{12} \\
&= \frac{(b-a+1)^2 - 1}{12}
\end{aligned}$$

□

- MGF:

$$M_X(t) = \frac{e^{at} - e^{(b+1)t}}{(b-a+1)(1-e^t)}$$

Proof.

$$E(e^{tX}) = \frac{1}{b-a+1} \sum_{k=a}^{b} e^{kt} = \frac{1}{b-a+1} e^{at} \frac{1 - e^{(b-a+1)t}}{1-e^t} = \frac{e^{at} - e^{(b+1)t}}{(b-a+1)(1-e^t)}$$

□

- Mode:

$$\text{Mode}(X) = \mathbb{Z} \cap [a, b]$$

- Median:

$$\text{Median}(X) = \begin{cases} \frac{a+b}{2}, & \frac{b-a}{2} \in \mathbb{Z} \\ \left[\frac{a+b-1}{2}, \frac{a+b+1}{2}\right], & \frac{b-a+1}{2} \in \mathbb{Z} \end{cases}$$

- Entropy:

$$H(X) = -\ln(b-a+1)$$

Proof.

$$\begin{aligned} H(X) &= -\sum_{k=a}^{b} \frac{1}{b-a+1} \ln\left(\frac{1}{b-a+1}\right) \\ &= -\ln(b-a+1) \end{aligned}$$

□

- PMF monotonicity: constant

ii Bernoulli (伯努力) random variable

- Trial: A Bernoulli random variable models a single Bernoulli trial (aka binary trial) with exactly two possible outcomes, success or 1, with probability p, and failure or 0, with probability $1-p$.
- Notation: $X \sim \text{Bern}(p)$, read as X is Bernoulli distributed with parameter p.
- Parameters: $p \in [0,1]$
- Support: $\{0,1\}$
- PMF:

$$p_X(k) = \begin{cases} p, & k=1 \\ 1-p, & k=0 \end{cases}$$

- CDF:

$$F_X(k) = \begin{cases} 0, & k<0 \\ 1-p, & 0 \le k < 1 \\ 1, & k \ge 1 \end{cases}$$

- Mean:

$$E(X) = p$$

- Variance:

$$\text{Var}(X) = p(1-p)$$

Proof.

$$E(X^2) = p$$

$$\text{Var}(X) = p - p^2 = p(1-p)$$

□

- MGF:

$$M_X(t) = pe^t + 1 - p$$

- Mode:

$$\text{Mode}(X) = \begin{cases} 0, & p < \frac{1}{2} \\ 0,1, & p = \frac{1}{2} \\ 1, & p > \frac{1}{2} \end{cases}$$

- Median:
$$\text{Median}(X) = \begin{cases} 0, & p < \frac{1}{2} \\ [0,1], & p = \frac{1}{2} \\ 1, & p > \frac{1}{2} \end{cases}$$

- Entropy:
$$H(X) = -p \ln p - (1-p)\ln(1-p)$$

- PMF monotonicity: strictly increasing if $p > \frac{1}{2}$, constant if $p = \frac{1}{2}$, and strictly decreasing if $p < \frac{1}{2}$

- Sum of independent random variables: Suppose $X_i \sim \text{Bern}(p)$ for $i \in \mathbb{N}_{\leq n}$ are independent random variables. Then
$$\sum_{i=1}^{n} X_i \sim \text{Bin}(n,p).$$

  Proof.
$$p_{\sum_{i=1}^{n} X_i}(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k \in \mathbb{Z} \cap [0,n]$$

  □

iii Binomial random variable

- Trial: A binomial random variable counts the number of successes in n Bernoulli trials with success probability p.

- Notation: $X \sim \text{Bin}(n,p)$, $X \sim B(n,p)$, $X \sim \text{Binom}(n,p)$, or $X \sim \text{Binomial}(n,p)$, read as X is binomial distributed with parameters n and p.

- Parameters: $n \in \mathbb{N}$, $p \in [0,1]$

- Support: $\mathbb{Z} \cap [0,n]$

- PMF:
$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k \in \mathbb{Z} \cap [0,n]$$

- CDF:
$$F_X(k) = \begin{cases} 0, & k < 0 \\ \displaystyle\sum_{i=0}^{\lfloor k \rfloor} \binom{n}{i} p^i (1-p)^{n-i}, & 0 \leq k < n \\ 1, & k \geq n \end{cases}$$

- Mean:
$$E(X) = np$$

Proof.

$$\begin{aligned}
E(X) &= \sum_{k=1}^{n} k\binom{n}{k} p^k(1-p)^{n-k} \\
&= \sum_{k=1}^{n} n\binom{n-1}{k-1} p^k(1-p)^{n-k} \\
&= np\sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1}(1-p)^{n-k} \\
&= np(p+1-p)^{n-1} \\
&= np
\end{aligned}$$

Alternatively, decompose it to n i.i.d. Bernoulli random variable with parameter p. Thus, the mean of X is n times that of the Bernoulli random variable. □

- Variance:

$$\text{Var}(X) = np(1-p)$$

Proof.

$$\begin{aligned}
E(X^2) &= \sum_{k=0}^{n} k^2\binom{n}{k} p^k(1-p)^{n-k} \\
&= np\sum_{k=1}^{n} k\binom{n-1}{k-1} p^{k-1}(1-p)^{n-k} \\
&= np\sum_{k=1}^{n-1} k\binom{n-1}{k} p^{k}(1-p)^{n-1-k} + np \\
&= np(n-1)p + np \\
&= n^2p^2 - np^2 + np
\end{aligned}$$

$$\text{Var}(X) = E(X^2) - E(X)^2 = -np^2 + np = np(1-p)$$

Alternatively, use decomposition. □

- MGF:

$$M_X(t) = \left(pe^t + 1 - p\right)^n$$

- Mode:

$$\text{Mode}(X) = \begin{cases} n, & p = 1 \\ (n+1)p - 1, (n+1)p, & (n+1)p \in \mathbb{N} \wedge p \notin \mathbb{Z} \\ \lfloor (n+1)p \rfloor, & \text{otherwise} \end{cases}$$

Proof. If $p = 0$, $p_X(k) = 0$. If $p = 1$, $p_X(k) = n$. Let

$$K \coloneqq \arg\max_k p_X(k)$$

Let

$$g(k) \coloneqq \frac{p_X(k+1)}{p_X(k)} = \frac{\binom{n}{k+1} p^{k+1}(1-p)^{n-k-1}}{\binom{n}{k} p^k(1-p)^{n-k}} = \frac{(n-k)p}{(k+1)(1-p)}$$

Extend $g(k)$ to the $\mathbb{R} \cap [0, n]$. For $k \in (0, n)$,

$$\begin{aligned} g'(k) &= \frac{-(k+1)-(n-k)}{(k+1)^2}\frac{p}{1-p} \\ &= \frac{-n+1}{(k+1)^2}\frac{p}{1-p} \\ &\leq 0 \end{aligned}$$

$$K = 1 + \min\{k \mid g(k) > 1\}, 1 + \min\{k \mid g(k) \geq 1\}$$

Solve $g(k*) = 1$ for $k*$.

$$\frac{n-k*}{k*+1} = \frac{1-p}{p}$$

$$np - k*p = k*+1 - k*p - p$$

$$k* = (n+1)p - 1$$

□

- PMF monotonicity: strictly increasing for $k \leq (n+1)p$ and strictly decreasing for $k \geq (n+1)p$
- Decomposition: sum of n i.i.d. Bernoulli random variables with parameter p
- Sum of independent random variables: Suppose $X_i \sim \text{Bin}(n_i, p)$ for $i \in \mathbb{N}_{\leq n}$ are independent random variables. Then

$$\sum_{i=1}^{n} X_i \sim \text{Bin}(\sum_{i=1}^{n} n_i, p).$$

Proof. Decompose to Bernoulli random variables. □

iv Geometric random variable

- Trial: A geometric random variable counts the number of independent Bernoulli trials with success probability p until the first success.
- Notation: $X \sim \text{Geom}(p)$, $X \sim G(p)$, or $X \sim \text{Geometric}(p)$, read as X is geometric distributed with parameter p.
- Parameters: $p \in (0, 1]$
- Support: $\mathbb{N}$
- PMF:

$$p_X(k) = p(1-p)^{k-1}, \quad k \in \mathbb{N}$$

- CDF:

$$F_X(k) = \begin{cases} 0, & k < 1 \\ 1 - (1-p)^{\lfloor k \rfloor}, & k \geq 1 \end{cases}$$

Proof.

$$
\begin{aligned}
P(X \leq k) &= \sum_{i=1}^{\lfloor k \rfloor} p(1-p)^{i-1} \\
&= \sum_{i=1}^{\lfloor k \rfloor} p(1-p)^{i-1} \\
&= \sum_{i=1}^{\infty} p(1-p)^{i-1} - \sum_{i=\lfloor k \rfloor+1}^{\infty} p(1-p)^{i-1} \\
&= 1 - (1-p)^{\lfloor k \rfloor} \sum_{i=1}^{\infty} p(1-p)^{i-1} \\
&= 1 - (1-p)^{\lfloor k \rfloor}
\end{aligned}
$$

□

- Mean:

$$E(X) = \frac{1}{p}$$

Proof.

$$
\begin{aligned}
E(X) &= \sum_{k=1}^{\infty} kp(1-p)^{k-1} \\
&= \sum_{k=1}^{\infty} (k-1)p(1-p)^{k-1} + \sum_{k=1}^{\infty} p(1-p)^{k-1} \\
&= \sum_{k=1}^{\infty} kp(1-p)^{k} + \sum_{k=1}^{\infty} p(1-p)^{k-1}
\end{aligned}
$$

$$(1-p)E(X) = \sum_{k=1}^{\infty} kp(1-p)^k$$

$$E(X) - (1-p)E(X) = pE(X) = \sum_{k=1}^{\infty} p(1-p)^{k-1} = \frac{p}{1-(1-p)} = 1$$

□

- Variance:

$$\text{Var}(X) = \frac{1-p}{p^2}$$

Proof.

$$
\begin{aligned}
E(X^2) &= \sum_{k=1}^{\infty} k^2 p(1-p)^{k-1} \\
&= \sum_{k=1}^{\infty} (k-1)^2 p(1-p)^{k-1} + 2\sum_{k=1}^{\infty} kp(1-p)^{k-1} - \sum_{k=1}^{\infty} p(1-p)^{k-1} \\
&= \sum_{k=1}^{\infty} k^2 p(1-p)^k + 2E(X) - 1
\end{aligned}
$$

$$(1-p)E(X^2) = \sum_{k=1}^{\infty} k^2 p(1-p)^k$$

$$E(X^2) - 2E(X) + 1 - (1-p)E(X^2) = pE(X^2) - 2E(X) + 1 = 0$$
$$E(X^2) = \frac{2E(X)-1}{p} = \frac{2-p}{p^2}$$
$$\text{Var}(X) = E(X^2) - E(X)^2 = \frac{1-p}{p^2}$$

□

- MGF:

$$M_X(t) = \frac{pe^t}{1-(1-p)e^t}, \quad t < -\ln(1-p)$$

Proof.

$$E(e^{tX}) = \sum_{k=1}^{\infty} e^{kt} p(1-p)^{k-1}$$
$$= \frac{pe^t}{1-(1-p)e^t}$$

□

- Mode:

$$\text{Mode}(X) = 1$$

Proof.

$$p_X(k+1) - p_X(k) = p(1-p)^k - p(1-p)^{k-1} = -p^2(1-p)^{k-1} < 0$$

□

- Median:

$$\text{Median}(X) = \begin{cases} \left[\frac{-\ln 2}{\ln(1-p)}, \frac{-\ln 2}{\ln(1-p)} + 1\right], & \frac{-\ln 2}{\ln(1-p)} \in \mathbb{N} \\ \left\lceil \frac{-\ln 2}{\ln(1-p)} \right\rceil, & \text{otherwise} \end{cases}$$

Proof.

$$P(X \le k) \ge \frac{1}{2}$$
$$1 - (1-p)^{\lfloor k \rfloor} \ge \frac{1}{2}$$
$$(1-p)^{\lfloor k \rfloor} \le \frac{1}{2}$$
$$\lfloor k \rfloor \ge \frac{-\ln 2}{\ln(1-p)}$$
$$P(X \ge k) \ge \frac{1}{2}$$
$$\sum_{i=\lceil k \rceil}^{\infty} p(1-p)^{i-1} = (1-p)^{\lceil k \rceil - 1} \sum_{i=1}^{\infty} p(1-p)^{i-1} = (1-p)^{\lceil k \rceil - 1} \ge \frac{1}{2}$$
$$\lceil k \rceil - 1 \le \frac{-\ln 2}{\ln(1-p)}$$

□

- Entropy:

$$H(X) = \frac{-p\ln p - (1-p)\ln(1-p)}{p}$$

Proof.

$$\begin{aligned}
H(X) &= -\sum_{k=1}^{\infty} p(1-p)^{k-1} \ln\left(p(1-p)^{k-1}\right) \\
&= -\sum_{k=1}^{\infty} p(1-p)^{k-1} \left(\ln p + (k-1)\ln(1-p)\right) \\
&= -\ln p - \ln(1-p)\sum_{k=1}^{\infty} kp(1-p)^{k-1} + \ln(1-p) \\
&= -\ln p - \ln(1-p)p\sum_{k=1}^{\infty} k(1-p)^{k-1} + \ln(1-p)
\end{aligned}$$

Let

$$\begin{aligned}
S &= \sum_{k=1}^{\infty} k(1-p)^{k-1} \\
&= \sum_{k=1}^{\infty}(k-1)(1-p)^{k-1} + \sum_{k=1}^{\infty}(1-p)^{k-1} \\
&= \sum_{k=1}^{\infty} k(1-p)^{k} + \frac{1}{p}
\end{aligned}$$

$$(1-p)S = \sum_{k=1}^{\infty} k(1-p)^{k}$$

$$S-(1-p)S = pS = \frac{1}{p}$$

$$S = \frac{1}{p^2}$$

$$H(X) = -\ln p - \ln(1-p)p\frac{1}{p^2} + \ln(1-p) = \frac{-p\ln p - (1-p)\ln(1-p)}{p}$$

□

- PMF monotonicity: strictly decreasing
- Memorylessness:

$$P(X = j+k \mid X > j) = P(X = k) \quad j,k \in \mathbb{N}$$

Proof.

$$\frac{p(1-p)^{j+k}}{(1-p)^{j}} = p(1-p)^{k}.$$

□

- Sum of independent random variables: Suppose $X_i \sim \text{Geom}(p)$ for $i \in \mathbb{N}_{\leq n}$ are independent random variables. Then

$$\sum_{i=1}^{n} X_i \sim \text{NegBin}(n,p).$$

Proof. Prove by mathematical induction. For $n = 1$, NegBin$(1, p)$ = Geom(p). Assume $\sum_{i=1}^{n-1} X_i \sim$ NegBin$(n-1, p)$, we show that $\sum_{i=1}^{n} X_i \sim$ NegBin(n, p).

$$
\begin{aligned}
p_{\sum_{i=1}^n X_i}(k) &= \sum_{i=n-1}^{k-1} \binom{i-1}{n-2} p^{n-1}(1-p)^{i-n+1} p(1-p)^{k-i-1} \\
&= p^n(1-p)^{k-n} \sum_{i=n-1}^{k-1} \binom{i-1}{n-2} = \qquad \binom{k-1}{n-1} p^n (1-p)^{k-n}
\end{aligned}
$$

□

v Negative binomial or Pascal random variable

- Trial: A negative binomial or Pascal random variable counts the number of independent Bernoulli trials with success probability p until the rth success.
- Notation: $X \sim$ NegBin(r, p), $X \sim$ Pascal(r, p), or $X \sim$ NB(r, p), read as X is negative binomial or Pascal distributed with parameters r and p.
- Parameters: $r \in \mathbb{N}$, $p \in (0, 1]$
- Support: $\mathbb{N} \cap [r, \infty)$
- PMF:
$$p_X(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k \in \mathbb{N} \cap [r, \infty)$$
- CDF:
$$F_X(k) = \begin{cases} 0, & k < r \\ \dfrac{p^r}{(r-1)!} \displaystyle\sum_{i=0}^{\lfloor k \rfloor - r} (i+r-1)_i (1-p)^i, & k \geq r \end{cases}$$
- Mean:
$$E(X) = \frac{r}{p}$$

  Proof. Consider a sequence of i.i.d. Bernoulli trial with success probability p with T_i defined as the random variable of the number of trials until the ith success for $i \in \mathbb{N} \cap [1, r]$. Define $X_1 = T_1$, and for $i \in \mathbb{N} \cap [2, r]$, define $X_i = T_i - T_{i-1}$. Then
$$X = T_r = \sum_{i=1}^{r} X_i.$$
  And the X_is are i.i.d. geometric random variable with parameter p. Thus, the mean of X is r times that of the geometric random variable. □
- Variance:
$$\text{Var}(X) = \frac{r(1-p)}{p^2}$$
- MGF:
$$M_X(t) = \left(\frac{pe^t}{1-(1-p)e^t} \right)^r, \quad t < -\ln(1-p)$$

- Mode:
$$\text{Mode}(X) = \left\lfloor \frac{r-1}{p} \right\rfloor$$

  Proof.
$$\begin{aligned} p_X(k+1) - p_X(k) &= \binom{k}{r-1} p^r (1-p)^{k-r+1} - \binom{k-1}{r-1} p^r (1-p)^{k-r} \\ &= (k(1-p) - (k-r+1)) \frac{(k-1)!}{(r-1)!(k-r+1)!} p^r (1-p)^{k-r} \\ &= (-kp + r - 1) \frac{(k-1)!}{(r-1)!(k-r+1)!} p^r (1-p)^{k-r} \end{aligned}$$

$$p_X(k+1) > p_X(k) \iff -kp + r - 1 > 0 \iff k < \frac{r-1}{p}$$
$$p_X(k+1) < p_X(k) \iff -kp + r - 1 < 0 \iff k > \frac{r-1}{p}$$

  □

- PMF monotonicity: strictly increasing for $k \leq \frac{r-1}{p}$ and strictly decreasing for $k \geq \frac{r-1}{p}$
- Decomposition: sum of r i.i.d. geometric random variable with parameter p
- Sum of independent random variables: Suppose $X_i \sim \text{NegBin}(r_i, p)$ for $i \in \mathbb{N}_{\leq n}$ are independent random variables. Then
$$\sum_{i=1}^{n} X_i \sim \text{NegBin}(\sum_{i=1}^{n} r_i, p).$$

  Proof. Decompose to geometric random variables. □

- Relation to binomial random variable: Suppose $X \sim \text{NegBin}(r, p)$ and $Y_n \sim \text{Bin}(n, p)$. Then, for all $n \in \mathbb{N} \cap [r, \infty)$,
$$F_X(n) + F_{Y_n}(r-1) = 1.$$

  Proof. $F_X(n)$ means that the rth success occurs before or at the nth trial, that is, there are at least r successes in the first n trials.

  $F_{Y_n}(r-1)$ means that there are at most $r-1$ successes in the first n trials.

  There are either at least r successes or at most $r-1$ successes in the first n trials. Thus their sum is one. □

vi Poisson random variable

- Trial: A Poisson random variable counts the number of events occurring in a fixed interval of time in a Poisson (point) process, in which events occur with a known constant mean rate λ and independently of the time since the last event.
- Notation: $X \sim \text{Pois}(\lambda)$ or $X \sim \text{Poisson}(\lambda)$, read as X is Poisson distributed with parameter or rate λ.
- Parameters: $\lambda \in (0, \infty)$

- Support: $\mathbb{N}_0$
- PMF:

$$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Proof. Let $N(t)$ denotes the number of events occuring in $[0, t]$, and let $p_k(t) = P(N(t) = k)$.

$$\lim_{h\to 0^+} p_0(h) = 1$$

$$\lim_{h\to 0^+} \frac{p_1(h)}{h} = \lambda$$

$$\lim_{h\to 0^+} \frac{p_k(h)}{h} = 0, \quad k \in \mathbb{Z} \cap [2, \infty)$$

$$\begin{aligned} p_k'(t) &= \lim_{h\to 0} \frac{p_k(t+h) - p_k(t)}{h} \\ &= \frac{p_k(t)(1-\lambda h) + p_{k-1}(t)(\lambda h) - p_k(t)}{h} \\ &= -\lambda p_k(t) + \lambda p_{k-1}(t) \end{aligned}$$

$$p_0'(t) = -\lambda p_0(t)$$

$$p_0(t) = e^{-\lambda t}$$

$$p_1'(t) = -\lambda p_1(t) + \lambda e^{-\lambda t}$$

$$\mu(t) = e^{\int \lambda \mathrm{d}t} = e^{\lambda t}$$

$$e^{\lambda t} p_1(t) = \lambda t$$

$$p_1(t) = \lambda t e^{-\lambda t}$$

Assume

$$p_k(t) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}.$$

Prove by mathematical induction. It holds for $k = 0$ and 1. Suppose it holds for $k \in \mathbb{Z} \cap [0, n]$. For $k = n + 1$,

$$p_{n+1}'(t) = -\lambda p_{n+1}(t) + \frac{\lambda(\lambda t)^n e^{-\lambda t}}{n!}$$

$$\mu(t) = e^{\int \lambda \mathrm{d}t} = e^{\lambda t}$$

$$e^{\lambda t} p_{n+1}(t) = \int \frac{\lambda(\lambda t)^n}{n!} \mathrm{d}t = \frac{(\lambda t)^{n+1}}{(n+1)!}$$

$$p_{n+1}(t) = \frac{(\lambda t)^{n+1} e^{-\lambda t}}{(n+1)!}$$

□

- CDF:

$$F_X(k) = e^{-\lambda} \sum_{i=0}^{\lfloor k \rfloor} \frac{\lambda^i}{i!}$$

- Mean:

$$E(X) = \lambda$$

Proof.

$$
\begin{aligned}
E(X) &= \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} \\
&= \lambda e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \\
&= \lambda e^{-\lambda} e^{\lambda} \\
&= \lambda
\end{aligned}
$$

□

- Variance:

$$\text{Var}(X) = \lambda$$

Proof.

$$
\begin{aligned}
E(X^2) &= \sum_{k=0}^{\infty} k^2 \frac{\lambda^k e^{-\lambda}}{k!} \\
&= \lambda e^{-\lambda} \sum_{k=0}^{\infty} (k+1) \frac{\lambda^k}{k!} \\
&= \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} + \lambda \\
&= \lambda^2 + \lambda \\
\text{Var}(X) &= E(X^2) - E(X)^2 = \lambda
\end{aligned}
$$

□

- MGF:

$$M_X(t) = \exp\left(\lambda \left(e^t - 1\right)\right)$$

Proof.

$$
\begin{aligned}
E(e^{tX}) &= \sum_{k=0}^{\infty} e^{kt} \frac{\lambda^k e^{-\lambda}}{k!} \\
&= e^{-\lambda} \sum_{k=0}^{\infty} \frac{\left(\lambda e^t\right)^k}{k!} \\
&= e^{-\lambda} \exp\left(\lambda e^t\right) \\
&= \exp\left(\lambda \left(e^t - 1\right)\right)
\end{aligned}
$$

□

- Mode:

$$\text{Mode}(X) = \lceil \lambda \rceil - 1, \lfloor \lambda \rfloor$$

Proof.

$$
\begin{aligned}
p_X(k+1) - p_X(k) &= \frac{\lambda^{k+1} e^{-\lambda}}{(k+1)!} - \frac{\lambda^k e^{-\lambda}}{k!} \\
&= (\lambda - k - 1) \frac{\lambda^k e^{-\lambda}}{(k+1)!} \\
p_X(k+1) > p_X(k) &\iff k < \lambda - 1
\end{aligned}
$$

$$p_X(k+1) < p_X(k) \iff k > \lambda - 1$$

$\square$

- Entropy:

$$H(X) = \lambda\left(1 - \ln\lambda\right) + e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k \ln(k!)}{k!}$$

Proof.

$$\begin{aligned}
H(X) &= -\sum_{k=0}^{\infty}\frac{\lambda^k e^{-\lambda}}{k!}\ln\left(\frac{\lambda^k e^{-\lambda}}{k!}\right) \\
&= -e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!}\left(k\ln\lambda - \lambda - \ln(k!)\right) \\
&= -e^{-\lambda}\ln\lambda\sum_{k=0}^{\infty}k\frac{\lambda^k}{k!} + e^{-\lambda}\lambda\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} + e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k\ln(k!)}{k!} \\
&= -e^{-\lambda}\lambda\ln\lambda\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} + \lambda + e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k\ln(k!)}{k!} \\
&= \lambda\left(1 - \ln\lambda\right) + e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k\ln(k!)}{k!}
\end{aligned}$$

$\square$

- PMF monotonicity: strictly increasing for $k \le \lambda - 1$ and strictly decreasing for $k \ge \lambda - 1$
- Addictivity: For any $n \in \mathbb{N}$, for any $X_i \sim \text{Pois}(\lambda_i)$ for all $i \in \mathbb{N} \cap [1, n]$,

$$\sum_{i=1}^{n} X_i \sim \text{Pois}\left(\sum_{i=1}^{n}\lambda_i\right)$$

Proof. Suppose $X \sim \text{Pois}(\lambda)$ and $Y \sim \text{Pois}(\rho)$.

$$\begin{aligned}
P(X_i + Y_i = k) &= \sum_{x=0}^{k} p_X(x)p_Y(k-x) \\
&= \sum_{x=0}^{k}\frac{\lambda^x e^{-\lambda}}{x!}\frac{\rho^{k-x}e^{-\rho}}{(k-x)!} \\
&= \frac{e^{-\lambda-\rho}}{k!}\sum_{x=0}^{k}\binom{k}{x}\lambda^x\rho^{k-x} \\
&= \frac{(\lambda+\rho)^k e^{-\lambda-\rho}}{k!}
\end{aligned}$$

$\square$

- Thinning property: Given $X \sim \text{Pois}(\lambda)$ for some $\lambda \in (0, \infty)$, a random variable Y such that $Y \mid (X = k) \sim \text{Bin}(k, p)$ for some $p \in (0, 1]$ must satisfy $Y \sim \text{Pois}(\lambda p)$.

Proof.

$$
\begin{aligned}
P(Y = m) &= \sum_{k=1}^{\infty} P(Y = m \mid X = k) P(X = k) \\
&= \sum_{k=m}^{\infty} \binom{k}{m} p^m (1-p)^{k-m} \frac{\lambda^k e^{-\lambda}}{k!} \qquad = \frac{(\lambda p)^m e^{-\lambda}}{m!} \sum_{k=m}^{\infty} \frac{(\lambda(1-p))^{k-m}}{(k-m)!} \\
&= \frac{(\lambda p)^m e^{-\lambda}}{m!} e^{\lambda(1-p)} \\
&= \frac{(\lambda p)^m e^{-\lambda p}}{m!}
\end{aligned}
$$

□

- Sum of independent random variables: Suppose $X_i \sim \text{Pois}(\lambda_i)$ for $i \in \mathbb{N}_{\leq n}$ are independent random variables. Then

$$
\sum_{i=1}^{n} X_i \sim \text{Pois}(\sum_{i=1}^{n} \lambda_i).
$$

Proof. Prove for two independent random variables $X \sim \text{Pois}(\mu)$ and $Y \sim \text{Pois}(\lambda)$.

$$
\begin{aligned}
p_{X+Y}(k) &= \sum_{j=0}^{k} \frac{\mu^j e^{-\mu}}{j!} \frac{\lambda^{k-j} e^{-\lambda}}{(k-j)!} \\
&= \frac{e^{-(\mu+\lambda)}}{k!} \sum_{j=0}^{k} \binom{k}{j} \mu^j \lambda^{k-j} \\
&= \frac{(\mu+\lambda)^k e^{-(\mu+\lambda)}}{k!}
\end{aligned}
$$

□

vii Binomial random variables converge in distribution to Poisson random variables theorem

For any fixed $\lambda \in (0, \infty)$, suppose $X \sim \text{Pois}(\lambda)$ and sequence $\left(X_n \sim \text{Bin}\left(n, \frac{\lambda}{n}\right)\right)_{n \in \mathbb{N}}$. Then

$$
X_n \xrightarrow{\text{d}} X \text{ as } n \to \infty.
$$

Proof.

$$
\begin{aligned}
\lim_{n\to\infty} P(X_n = k) &= \lim_{n\to\infty} \binom{n}{k} \frac{\lambda^k}{n^k} \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
&= \lim_{n\to\infty} \frac{n^k}{k!} \frac{\lambda^k}{n^k} e^{-\lambda} \left(1 - \frac{\lambda}{n}\right)^{-k} \\
&= \lim_{n\to\infty} \frac{\lambda^k}{k!} e^{-\lambda} \left(1 - \frac{\lambda}{n}\right)^{-k} \\
&= \frac{\lambda^k e^{-\lambda}}{k!}
\end{aligned}
$$

□

XXVI Families of Absolutely Continuous Random Variables

i Continuous or rectangular uniform random variable

- Notation: $X \sim \mathcal{U}_{[a,b]}$, $X \sim U_{[a,b]}$, $X \sim$ Unif(a, b), $X \sim$ Unif$[a, b]$, $X \sim$ Uniform(a, b), or $X \sim$ Uniform$[a, b]$, read as X is continuously or rectangularly uniform distributed with parameters a and b.
- Parameters: $a, b \in \mathbb{R} \wedge a < b$
- Support: $[a, b]$
- PDF:
$$f_X(x) = \frac{1}{b-a}, \quad x \in [a, b]$$
- CDF:
$$F_X(x) = \begin{cases} 0, & x < a \\ \dfrac{x-a}{b-a}, & x \in [a, b] \\ 1, & x > b \end{cases}$$
- Mean:
$$E(X) = \frac{a+b}{2}$$
- Variance:
$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

  Proof.
$$\begin{aligned} E(X^2) &= \int_a^b \frac{x^2}{b-a}\,\mathrm{d}x \\ &= \frac{1}{b-a}\left(\frac{x^3}{3}\right)\Bigg|_a^b \\ &= \frac{a^2+b^2+ab}{3} \end{aligned}$$
$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 \\ &= \frac{a^2+b^2+ab}{3} - \frac{a^2+b^2+2ab}{4} \\ &= \frac{4a^2+4b^2+4ab-3a^2-3b^2-6ab}{12} \\ &= \frac{(b-a)^2}{12} \end{aligned}$$
$\square$

- MGF:
$$M_X(t) = \begin{cases} \dfrac{e^{bt}-e^{at}}{(b-a)t}, & t \neq 0 \\ 1, & t = 0 \end{cases}$$

Proof.

$$\begin{aligned} E(e^{tX}) &= \int_a^b \frac{e^{tx}}{b-a}\,\mathrm{d}x \\ &= \begin{cases} \dfrac{e^{bt}-e^{at}}{(b-a)t}, & t \neq 0 \\ 1, & t = 0 \end{cases} \end{aligned}$$

□

- Mode:

$$\text{Mode}(X) = [a, b]$$

- Median:

$$\text{Median}(X) = \frac{a+b}{2}$$

- Entropy:

$$h(X) = \ln(b-a)$$

Proof.

$$\begin{aligned} h(X) &= -\int_a^b \frac{1}{b-a} \ln\left(\frac{1}{b-a}\right)\mathrm{d}x \\ &= \ln(b-a) \end{aligned}$$

□

- Transition:

$$X - c \sim \text{Unif}(a-c, b-c), \quad c \in \mathbb{R}$$

- Scaling:

$$cX \sim \text{Unif}(ca, cb), \quad c \in \mathbb{R}_{>0}$$

$$cX \sim \text{Unif}(cb, ca), \quad c \in \mathbb{R}_{<0}$$

- Symmetry: about $\dfrac{b-a}{2}$
- PDF monotonicity: constant
- Floor:

$$\lfloor X \rfloor \sim \mathcal{U}\{a, b-1\}.$$

- Ceiling:

$$\lceil X \rceil \sim \mathcal{U}\{a+1, b\}.$$

ii Buffon's needle problem

Consider a surface ruled with parallel lines, where the distance between adjacent line is d. Suppose that we drop a needle of length $l < d$ on the surface at random. Then, the probability that the needle will intersect one of the lines is $\dfrac{2l}{\pi d}$.

Proof. Let X be the vertical distance from the midpoint of the needle to the nearest of the parallel lines, and Θ the acute angle formed by the axis of the needle and the parallel lines.

$$X \sim \text{Uniform}\left(0, \frac{d}{2}\right)$$

$$\Theta \sim \text{Uniform}\left(0, \frac{\pi}{2}\right)$$

$$f_X(x) = \begin{cases} \frac{2}{d}, & 0 \le x \le \frac{d}{2} \\ 0, & \text{otherwise} \end{cases}$$

$$f_\Theta(\theta) = \begin{cases} \frac{2}{\pi}, & 0 \le x \le \frac{\pi}{2} \\ 0, & \text{otherwise} \end{cases}$$

The event that the needle intersects one of the lines is $X \le \frac{l}{2} \sin \Theta$.

$$\begin{aligned} P(X \le \frac{l}{2} \sin \Theta) &= \int_0^{\frac{\pi}{2}} P(X \le \frac{l}{2} \sin \theta) \frac{2}{\pi} \, \mathrm{d}\theta \\ &= \int_0^{\frac{\pi}{2}} \frac{l}{2} \sin \theta \frac{2}{d} \frac{2}{\pi} \, \mathrm{d}\theta \\ &= \frac{2l}{\pi d} \left(-\cos \theta \right) \Big|_0^{\frac{\pi}{2}} \\ &= \frac{2l}{\pi d} \end{aligned}$$

□

iii Normal (常態/正態) or Gaussian (高斯) random variable

- Notation:
  - $X \sim \mathcal{N}(\mu, \sigma^2)$, $X \sim N(\mu, \sigma^2)$, or $X \sim \text{Gaussian}(\mu, \sigma^2)$, read as X is normal or Gaussian distributed with mean μ and standard deviation σ or variance σ^2.
  - $Z \sim \mathcal{N}(0, 1)$, $Z \sim N(0, 1)$, or $X \sim \text{Gaussian}(0, 1)$, read as X is standard or unit normal or Gaussian distributed.
- Parameters: $\mu \in \mathbb{R}$ and $\sigma \in (0, \infty)$
- Support: $\mathbb{R}$
- PDF:

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} \right) \\ &= \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma} \right), \end{aligned}$$

  where $\varphi(x)$ is the PDF of Z, that is,

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2} \right)$$

- CDF:
$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right),$$
where $\Phi(x)$ is the CDF of Z, that is,
$$\Phi(x) = \frac{1}{2}\left(1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right)\right),$$
where erf(x) is the (Gaussian) error function, that is,
$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,\mathrm{d}t$$

Proof. Lemma:
$$I = \int_0^\infty e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}$$

Proof of lemma:
$$\begin{aligned}
I^2 &= \left(\int_0^\infty e^{-x^2}\,\mathrm{d}x\right)\left(\int_0^\infty e^{-y^2}\,\mathrm{d}y\right) \\
&= \int_0^\infty \int_0^\infty e^{-x^2} e^{-y^2}\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_0^{\pi/2} \int_0^\infty e^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \frac{1}{2}\int_0^{\pi/2} \int_0^\infty e^{-u}\,\mathrm{d}u\,\mathrm{d}\theta\,, \quad u = r^2 \\
&= \frac{1}{2}\int_0^{\pi/2} 1\,\mathrm{d}\theta \\
&= \frac{\pi}{4}
\end{aligned}$$

$$F_X(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right)\mathrm{d}t$$

$$u = \frac{t-\mu}{\sqrt{2}\sigma}, \quad \mathrm{d}u = \frac{1}{\sqrt{2}\sigma}\,\mathrm{d}t$$

$$\begin{aligned}
F_X(x) &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\frac{x-\mu}{\sqrt{2}\sigma}} e^{-u^2}\,\mathrm{d}u \\
&= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\frac{x-\mu}{\sqrt{2}\sigma}} e^{-u^2}\,\mathrm{d}u \\
&= \frac{1}{\sqrt{\pi}}\left(\frac{\sqrt{\pi}}{2} + \int_0^{\frac{x-\mu}{\sqrt{2}\sigma}} e^{-u^2}\,\mathrm{d}u\right) \\
&= \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{\frac{x-\mu}{\sqrt{2}\sigma}} e^{-u^2}\,\mathrm{d}u \\
&= \frac{1}{2}\left(1 + \text{erf}\left(\frac{x-\mu}{\sqrt{2}\sigma}\right)\right)
\end{aligned}$$

$\square$

- Quantile:

$$\begin{aligned} Q_X(p) &= \mu + \sigma\Phi^{-1}(p) \\ &= \mu + \sqrt{2}\sigma\,\mathrm{erf}^{-1}(2p-1) \end{aligned}$$

- Mean:

$$E(X) = \mu$$

Proof.

$$E(X) = \int_{-\infty}^{\infty} \frac{x}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \mathrm{d}x$$

$$u = \frac{x-\mu}{\sqrt{2}\sigma}, \quad \mathrm{d}u = \frac{1}{\sqrt{2}\sigma}\,\mathrm{d}x$$

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} \frac{x}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \mathrm{d}x \\ &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \left(\sqrt{2}\sigma u + \mu\right) e^{-u^2}\,\mathrm{d}u \\ &= \mu \end{aligned}$$

□

- Variance:

$$\mathrm{Var}(X) = \sigma^2$$

Proof.

$$E(X^2) = \int_{-\infty}^{\infty} \frac{x^2}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \mathrm{d}x$$

$$u = \frac{x-\mu}{\sqrt{2}\sigma}, \quad \mathrm{d}u = \frac{1}{\sqrt{2}\sigma}\,\mathrm{d}x$$

$$\begin{aligned} E(X^2) &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \left(\sqrt{2}\sigma u + \mu\right)^2 e^{-u^2}\,\mathrm{d}u \\ &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} 2\sigma^2 u^2 e^{-u^2}\,\mathrm{d}u + \mu^2 \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} u^2 e^{-u^2}\,\mathrm{d}u + \mu^2 \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \left(\left(-\frac{1}{2}ue^{-u^2}\right)\Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} ue^{-u^2}\,\mathrm{d}u\right) + \mu^2 \\ &= -\frac{2\sigma^2}{\sqrt{\pi}} \left(\left(-\frac{1}{2}e^{-u^2}\right)\Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{1}{2}e^{-u^2}\,\mathrm{d}u\right) + \mu^2 \\ &= \sigma^2 + \mu^2 \end{aligned}$$

$$\mathrm{Var}(X) = E(X^2) - E(X)^2 = \sigma^2 + \mu^2 - \mu^2 = \sigma^2$$

□

- MGF:

$$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$$

Proof.

$$
\begin{aligned}
E(e^{tX}) &= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty}\operatorname{exp}\left(-\frac{(x-\mu)^2}{2\sigma^2}+tx\right)\mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty}\operatorname{exp}\left(-\frac{(x-\mu-\sigma^2 t)^2}{2\sigma^2}+\frac{\sigma^2 t^2}{2}+\mu t\right)\mathrm{d}x \\
&= \frac{\operatorname{exp}\left(\frac{\sigma^2 t^2}{2}+\mu t\right)}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty}\operatorname{exp}\left(-\frac{(x-\mu-\sigma^2 t)^2}{2\sigma^2}\right)\mathrm{d}x \\
&= \operatorname{exp}\left(\mu t+\frac{\sigma^2 t^2}{2}\right)
\end{aligned}
$$

□

- Mode:

$$
\operatorname{Mode}(X) = \mu
$$

Proof.

$$
\begin{aligned}
f_X'(x) &= -\frac{1}{\sqrt{2\pi}\sigma}\frac{2(x-\mu)}{2\sigma^2}\operatorname{exp}\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\
&= -\frac{(x-\mu)}{\sqrt{2\pi}\sigma^3}\operatorname{exp}\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\
f_X'(x) > 0 &\iff x < \mu \\
f_X'(x) < 0 &\iff x > \mu
\end{aligned}
$$

□

- Median:

$$
\operatorname{Median}(X) = \mu
$$

- Entropy:

$$
h(X) = \frac{1}{2}\ln\left(2\pi e\sigma^2\right)
$$

Proof.

$$
u = \frac{x-\mu}{\sqrt{2}\sigma}, \quad \mathrm{d}u = \frac{1}{\sqrt{2}\sigma}\,\mathrm{d}x
$$

$$
\begin{aligned}
h(X) &= -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \operatorname{exp}\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \ln\left(\frac{1}{\sqrt{2\pi}\sigma} \operatorname{exp}\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\right) \mathrm{d}x \\
&= -\frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \operatorname{exp}\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \left(\ln\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{(x-\mu)^2}{2\sigma^2}\right) \mathrm{d}x \\
&= \frac{\ln\left(\sqrt{2\pi}\sigma\right)}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \operatorname{exp}\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \mathrm{d}x + \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \operatorname{exp}\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \frac{(x-\mu)^2}{2\sigma^2} \mathrm{d}x \\
&= \frac{\ln\left(2\pi\sigma^2\right)}{2\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-u^2} \mathrm{d}u + \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} u^2 e^{-u^2} \mathrm{d}u \\
&= \frac{\ln\left(2\pi\sigma^2\right)}{2} + \frac{1}{\sqrt{\pi}} \left(\left(-\frac{1}{2} u e^{-u^2}\right)\Big|_{-\infty}^{\infty} + \frac{1}{2} \int_{-\infty}^{\infty} e^{-u^2} \mathrm{d}x\right) \\
&= \frac{\ln\left(2\pi\sigma^2\right)}{2} + \frac{1}{2} \\
&= \frac{1}{2} \ln\left(2\pi e \sigma^2\right)
\end{aligned}
$$

□

- Standardization:

$$X = \mu + \sigma Z$$

- Transition:

$$X - c \sim \mathcal{N}(\mu - c, \sigma^2), \quad c \in \mathbb{R}$$

- Scaling:

$$cX \sim \mathcal{N}(c\mu, c^2\sigma^2), \quad c \in \mathbb{R}_{\neq 0}$$

- Symmetry: about μ
- PDF monotonicity: strictly increasing for $x \leq \mu$ and strictly decreasing for $x \geq \mu$
- CCDF:

$$\bar{F}_X(x) = F_X(-x)$$

Proof.

$$
\begin{aligned}
\Phi(-z) + \Phi(z) &= \int_{-\infty}^{-z} \frac{1}{\sqrt{2\pi}} \operatorname{exp}\left(-\frac{x^2}{2}\right) \mathrm{d}x + \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \operatorname{exp}\left(-\frac{x^2}{2}\right) \mathrm{d}x \\
&= \int_{z}^{\infty} \frac{1}{\sqrt{2\pi}} \operatorname{exp}\left(-\frac{x^2}{2}\right) \mathrm{d}x + \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \operatorname{exp}\left(-\frac{x^2}{2}\right) \mathrm{d}x \\
&= 1
\end{aligned}
$$

□

The CCDF of Z is called Q-function and denoted as $Q(x)$.

- Sum of independent random variables: Suppose $X_i \sim \mathcal{N}(\mu_i, {\sigma_i}^2)$ for $i \in \mathbb{N}_{\leq n}$ are independent random variables. Then

$$\sum_{i=1}^{n} X_i \sim \mathcal{N}(\sum_{i=1}^{n} \mu_i, \sum_{i=1}^{n} {\sigma_i}^2).$$

Proof. Prove for two independent random variables $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y \sim \mathcal{N}(\lambda, \gamma^2)$.

$$
\begin{aligned}
f_{X+Y}(z) = &\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \frac{1}{\sqrt{2\pi}\gamma} \exp\left(-\frac{(z-x-\lambda)^2}{2\gamma^2}\right) \mathrm{d}x \\
=&\frac{1}{2\pi\sigma\gamma} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}\left(\frac{(x-\mu)^2}{\sigma^2} + \frac{(x-(z-\lambda))^2}{\gamma^2}\right)\right) \mathrm{d}x \\
=&\frac{1}{2\pi\sigma\gamma} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}\left(x^2\left(\frac{1}{\sigma^2}+\frac{1}{\gamma^2}\right) - 2x\left(\frac{\mu}{\sigma^2}+\frac{z-\lambda}{\gamma^2}\right) + \left(\frac{\mu^2}{\sigma^2}+\frac{(z-\lambda)^2}{\gamma^2}\right)\right)\right) \mathrm{d}x\,, \quad \alpha = \frac{1}{\sigma^2} \\
=&\frac{1}{2\pi\sigma\gamma} \exp\left(\frac{1}{2}\left(\frac{\beta^2}{\alpha} - C(z)\right)\right) \int_{-\infty}^{\infty} \exp\left(-\frac{\alpha}{2}\left(x - \frac{\beta}{\alpha}\right)^2\right) \mathrm{d}x\,, \quad u = \frac{\sqrt{\alpha}}{\sqrt{2}}\left(x - \frac{\beta}{\alpha}\right) \\
=&\frac{1}{\pi\sqrt{2\alpha}\sigma\gamma} \exp\left(\frac{1}{2}\left(\frac{\beta^2}{\alpha} - C(z)\right)\right) \int_{-\infty}^{\infty} \exp(-u^2)\, \mathrm{d}u \\
=&\frac{1}{\sqrt{2\pi\left(\sigma^2+\gamma^2\right)}} \exp\left(\frac{1}{2}\left(\frac{\left(\mu\gamma^2 + (z-\lambda)\,\sigma^2\right)^2}{\sigma^2\gamma^2\left(\sigma^2+\gamma^2\right)} - \frac{\mu^2}{\sigma^2} - \frac{(z-\lambda)^2}{\gamma^2}\right)\right) \\
=&\frac{1}{\sqrt{2\pi\left(\sigma^2+\gamma^2\right)}} \exp\left(\frac{1}{2}\frac{\mu^2\gamma^4 + 2z\mu\sigma^2\gamma^2 - 2\mu\lambda\sigma^2\gamma^2 + z^2\sigma^4 - 2z\lambda\sigma^4 + \lambda^2\sigma^4 - \mu^2\sigma^2\gamma^2 - \mu^2\gamma^4 - z}{\sigma^2\gamma^2\left(\sigma^2+\gamma^2\right)}\right. \\
=&\frac{1}{\sqrt{2\pi\left(\sigma^2+\gamma^2\right)}} \exp\left(-\frac{(z-(\mu+\lambda))^2}{2\left(\sigma^2+\gamma^2\right)}\right)
\end{aligned}
$$

□

iv Q-function bounds and limit

$$\frac{x}{1+x^2}\varphi(x) < Q(x) < \frac{\varphi(x)}{x}$$

Proof.

$$
\begin{aligned}
Q(x) = &\int_x^{\infty} \varphi(u)\, \mathrm{d}u \\
< &\int_x^{\infty} \frac{u}{x}\varphi(u)\, \mathrm{d}u \\
=&\frac{1}{\sqrt{2\pi}} \int_x^{\infty} \frac{u}{x} e^{-\frac{x^2}{2}}\, \mathrm{d}u\,, \quad v = \frac{u^2}{2} \\
=&\frac{1}{\sqrt{2\pi}} \int_x^{\infty} \frac{u}{x} \exp\left(-\frac{u^2}{2}\right) \mathrm{d}u\,, \quad v = \frac{u^2}{2}, \mathrm{d}v = u\, \mathrm{d}u \\
=&\frac{1}{x\sqrt{2\pi}} \int_{\frac{x^2}{2}}^{\infty} e^{-v}\, \mathrm{d}v \\
=&\frac{1}{x\sqrt{2\pi}} \left(-e^{-v}\right)\Big|_{\frac{x^2}{2}}^{\infty} \\
=&\frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}} \\
=&\frac{\varphi(x)}{x}
\end{aligned}
$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}u}\left(\frac{\varphi(u)}{u}\right) =& \frac{-u\varphi(u)u - \varphi(u)}{u^2} \\
=& -\left(1 + \frac{1}{u^2}\right)\varphi(u) \\
\left(1 + \frac{1}{x^2}\right)Q(x) =& \int_x^\infty \left(1 + \frac{1}{x^2}\right)\varphi(u)\,\mathrm{d}u \\
>& \int_x^\infty \left(1 + \frac{1}{u^2}\right)\varphi(u)\,\mathrm{d}u \\
=& \left(-\frac{\varphi(u)}{u}\right)\Bigg|_x^\infty \\
=& \frac{\varphi(x)}{x} \\
Q(x) >& \frac{\varphi(x)}{x}\left(1 + \frac{1}{x^2}\right)^{-1} \\
=& \frac{x}{1+x^2}\varphi(x)
\end{aligned}$$

□

$$\lim_{x\to\infty} \frac{Q(x)}{\varphi(x)/x} = 1$$

Proof.

$$\lim_{x\to\infty} \frac{x^2}{1+x^2} = 1$$

□

v Stein's Lemma

Suppose $X \sim N(\mu, \sigma)$ and $g : \mathbb{R} \to \mathbb{R}$ is a differentiable function, and both $E(g(X)(X-\mu))$ and $E(g'(X))$ both exist. Then:

$$E(g(X)(X-\mu)) = \sigma^2 E(g'(X)).$$

Proof.

$$\begin{aligned}
E(g(X)(X-\mu)) =& \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} g(x)(x-\mu)\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\mathrm{d}x \\
=& \frac{\sigma}{\sqrt{2\pi}}\int_{-\infty}^{\infty} -g(x)\frac{x-\mu}{\sigma^2}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\mathrm{d}x \\
=& \frac{\sigma}{\sqrt{2\pi}}\int_{-\infty}^{\infty} g'(x)\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\mathrm{d}x \\
=& \sigma^2 E(g'(X))
\end{aligned}$$

□

vi Multidimensional Stein's Lemma

Suppose $X_i \sim N(\mu_i, \sigma)$ for $i \in \mathbb{Z} \cap [1, n]$ are independent, $\mathbf{X} = (X_1, \ldots, X_n)$, $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_n)$, and $f : \mathbb{R}^n \to \mathbb{R}^m$ is a differentiable function, and both $E((\mathbf{X} - \boldsymbol{\mu}) f(\mathbf{X})^\top)$ and $E(\nabla f(\mathbf{X})^\top)$ both exist. Then:

$$E((\mathbf{X} - \boldsymbol{\mu}) f(\mathbf{X})^\top) = \sigma^2 E(\nabla f(\mathbf{X})^\top).$$

Proof. By independency, apply Stein's lemma for each entry. □

vii Student's t-distribution, Student's t distribution, t-distribution, or t distribution

- Notation: $X \sim t_\nu$, read as X is student's t distributed with degrees of freedom ν.
- Parameters: $\nu \in \mathbb{N}$
- Support: $\mathbb{R}$
- PDF:

$$f_X(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}},$$

  where $\Gamma(x)$ is the Gamma function:

$$\Gamma(x) = \int_0^\infty u^{x-1} e^{-u}\, \mathrm{d}u\,, \quad x > 0.$$

- Mean:

$$\text{Mean}(X) = \begin{cases} 0, & \nu > 1, \\ \text{undefined}, & \text{otherwise}. \end{cases}$$

  Proof. PLACEHOLDER □

- Variance:

$$\text{Var}(x) = \begin{cases} \frac{\nu}{\nu - 2}, & \nu > 2, \\ \infty, & 1 < \nu \leq 2, \\ \text{undefined}, & \text{otherwise}. \end{cases}$$

  Proof. PLACEHOLDER □

- MGF:

$$M_X(t) = \text{undefined}.$$

  Proof. PLACEHOLDER □

- Mode:

$$\text{Mode}(X) = 0$$

  Proof. PLACEHOLDER □

- Median:

$$\text{Median}(X) = 0$$

  Proof. PLACEHOLDER □

- Symmetry: about 0

  Proof. PLACEHOLDER $\square$

- PDF monotonicity: strictly increasing for $x \le 0$ and strictly decreasing for $x \ge 0$

  Proof. PLACEHOLDER $\square$

- Relation to standard normal random variable: Suppose $Z_i \sim \mathcal{N}(0,1)$ for $i \in \mathbb{Z} \cap [0, \nu]$ are independent random variables.

$$X = \frac{Z_0}{\sqrt{\frac{1}{\nu}\sum_{i=1}^{\nu} {Z_i}^2}}.$$

  Proof. PLACEHOLDER $\square$

viii Student's t-distributions converge in distribution to standard normal distribution theorem

Let $(t_\nu)_{i=1}^{\infty}$ be a sequence and $Z \sim \mathcal{N}(0,1)$. Then

$$t_\nu \xrightarrow{\text{d}} Z \text{ as } \nu \to \infty.$$

Proof. PLACEHOLDER $\square$

ix Exponential random variable

- Trial: An exponential random variable models the distance between adjacent events in a Poisson process with rate λ.
- Notation: $X \sim \text{Exp}(\lambda)$ or $X \sim \text{Exponential}(\lambda)$, read as X is exponential distributed with parameter or rate λ.
- Parameters: $\lambda \in (0, \infty)$
- Support: $[0, \infty)$
- PDF:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \in \mathbb{R}_{\ge 0}$$

  Proof. Let $N(x)$ denotes the number of events occuring in $[0, x]$. We already know that

$$P(N(x) = k) = \frac{(\lambda x)^k e^{-\lambda t}}{k!}.$$

$$X = \inf\{x > 0 \mid N(x) \ge 1\}$$

$$\{X > x\} \iff \{N(x) = 0\}.$$

$$P(X > x) = P(N(x) = 0) = e^{-\lambda x}$$

$$F_X(x) = P(X \le x) = 1 - e^{-\lambda x}$$

$$f_X(x) = \lambda e^{-\lambda x}$$

  $\square$

- CDF:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0 \end{cases}$$

- Quantile:

$$Q_X(p) = -\frac{\ln(1-p)}{\lambda}$$

- Mean:

$$E(X) = \frac{1}{\lambda}$$

Proof.

$$\begin{aligned} E(X) &= \int_0^\infty x\lambda e^{-\lambda x}\,\mathrm{d}x \\ &= \left(-xe^{-\lambda x}\right)\Big|_0^\infty + \int_0^\infty e^{-\lambda x}\,\mathrm{d}x \\ &= \left(-\frac{1}{\lambda}e^{-\lambda x}\right)\Big|_0^\infty \\ &= \frac{1}{\lambda} \end{aligned}$$

□

- Variance:

$$\mathrm{Var}(X) = \frac{1}{\lambda^2}$$

Proof.

$$\begin{aligned} E(X^2) &= \int_0^\infty x^2\lambda e^{-\lambda x}\,\mathrm{d}x \\ &= \left(-x^2e^{-\lambda x}\right)\Big|_0^\infty + \int_0^\infty 2xe^{-\lambda x}\,\mathrm{d}x \\ &= \frac{2}{\lambda^2} \end{aligned}$$

$$\mathrm{Var}(X) = E(X^2) - E(X)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

□

- MGF:

$$M_X(t) = \left(1 - \frac{t}{\lambda}\right)^{-1}, \quad t < \lambda$$

Proof.

$$\begin{aligned} E(e^{tX}) &= \int_0^\infty \lambda e^{(t-\lambda)x}\,\mathrm{d}x \\ &= \left(\frac{\lambda e^{(t-\lambda)x}}{t-\lambda}\right)\Big|_0^\infty \\ &= \frac{\lambda}{\lambda - t} \\ &= \left(1 - \frac{t}{\lambda}\right)^{-1} \end{aligned}$$

□

- Mode:
$$\text{Mode}(X) = 0$$

  Proof.
$$f_X'(x) = -\lambda^2 e^{-\lambda x} < 0$$
$\square$

- Median:
$$\text{Median}(X) = \frac{\ln 2}{\lambda}$$

- Entropy:
$$h(X) = 1 - \ln \lambda$$

  Proof.
$$\begin{aligned} h(X) &= -\lambda \int_0^\infty e^{-\lambda x} \left(\ln(\lambda) - \lambda x\right) \mathrm{d}x \\ &= \left(e^{-\lambda x} \ln(\lambda)\right)\Big|_0^\infty + \lambda^2 \int_0^\infty x e^{-\lambda x} \mathrm{d}x \\ &= 1 - \ln \lambda \end{aligned}$$
$\square$

- Scaling:
$$cX \sim \text{Exp}\left(\frac{\lambda}{c}\right), \quad c \in \mathbb{R}_{>0}$$

- Memorylessness:
$$P(X > x + y | X > x) = P(X > y), \quad x, y \in \mathbb{R}_{\geq 0}$$

  Proof.
$$\frac{e^{-\lambda(x+y)}}{e^{-\lambda x}} = e^{-\lambda y}$$
$\square$

- PDF monotonicity: strictly decreasing

- Relation to Poisson random variable: For $\lambda, x > 0$,
$$P(\text{Exp}(\lambda) > x) = P(\text{Pois}(\lambda x) = 0),$$
  equivalently,
$$P(\text{Exp}(\lambda) \leq x) = P(\text{Pois}(\lambda x) \geq 1).$$

  Proof.
$$\begin{aligned} P(\text{Exp}(\lambda) > x) &= 1 - F_X(x) \\ &= e^{-\lambda x} \\ &= \frac{\lambda^0 e^{-\lambda}}{0!} \end{aligned}$$
$\square$

- Ceiling:
$$\lceil X \rceil \sim \text{Geom}(1 - e^{-\lambda}).$$

Proof.

$$\begin{aligned}P(k-1 < x \le k) = &\int_{k-1}^{k} \lambda e^{-\lambda x}\,\mathrm{d}x \\ =&\left(-e^{-\lambda x}\right)\Big|_{k-1}^{k} \\ =&e^{-\lambda(k-1)} - e^{-\lambda k} \\ =&e^{-\lambda(k-1)}\left(1-e^{-\lambda}\right) \\ =&\left(1-e^{-\lambda}\right)\left(e^{-\lambda}\right)^{k-1}\end{aligned}$$

□

- CCDF:

$$\bar{F}_X(x) = e^{-\lambda x}.$$

- Rates add under minimum: Suppose $X_i \sim \text{Exp}(\lambda_i)$ for $i \in \mathbb{N}_{\le n}$ are independent random variables.

$$\min_{i\in\mathbb{N}_{\le n}} X_i \sim \text{Exp}(\sum_{i\in I} \lambda_i).$$

Proof.

$$P(\min_{i\in\mathbb{N}_{\le n}} X_i > x) = \prod_{i=1}^{n} e^{-\lambda_i x} = e^{-\sum_{i=1}^{n} \lambda_i x}$$

□

- Sum of independent random variables: Suppose $X_i \sim \text{Exp}(\lambda)$ for $i \in \mathbb{N}_{\le n}$ are independent random variables. Then

$$\sum_{i=1}^{n} X_i \sim \text{Erlang}(n, \lambda).$$

Proof. Prove by mathematical induction. For $n = 1$, $\text{Erlang}(1, \lambda) = \text{Exp}(\lambda)$. Assume $\sum_{i=1}^{n-1} X_i \sim \text{Erlang}(n-1, \lambda)$, we show that $\sum_{i=1}^{n} X_i \sim \text{Erlang}(n, \lambda)$.

$$\begin{aligned}f_{\sum_{i=1}^{n} X_i}(z) = &\int_0^{\infty} \frac{\lambda^k x^{k-1} e^{-\lambda x}}{(k-1)!} \lambda e^{-\lambda(z-x)}\,\mathrm{d}x \\ =&\int_0^{\infty} \frac{\lambda^{k+1} x^{k-1} e^{-\lambda z}}{(k-1)!}\,\mathrm{d}x \\ =&\frac{\lambda^{k+1} x^k e^{-\lambda z}}{k!}\end{aligned}$$

□

x Erlang random variable

- Trial/Decomposition: An random variable with shape k and rate λ is the sum of k i.i.d. exponential variables with rate λ.
- Notation: $X \sim \text{Erlang}(k, \lambda)$, read as X is Erlang distributed with shape k and rate λ, or parameters k and λ.

- Parameters: $k \in \mathbb{N}$ and $\lambda \in (0, \infty)$
- Support: $[0, \infty)$
- PDF:

$$f_X(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{(k-1)!}, \quad x \in \mathbb{R}_{\geq 0}$$

Proof. Prove by mathematical induction. For $k = 1$,

$$f_1(x) = \lambda e^{-\lambda x}$$

Assume for $k = n$,

$$f_n(x) = \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!}$$

For $k = n + 1$,

$$\begin{aligned}
f_{n+1}(x) &= \int_0^x f_n(t) f_1(x-t)\,\mathrm{d}t \\
&= \int_0^x \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!} \lambda e^{-\lambda(x-t)}\,\mathrm{d}t \\
&= \int_0^x \frac{\lambda^{n+1} t^{n-1} e^{-\lambda x}}{(n-1)!}\,\mathrm{d}t \\
&= \frac{\lambda^{n+1} x^n e^{-\lambda x}}{n!}
\end{aligned}$$

□

- CDF:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - \displaystyle\sum_{n=0}^{k-1} \frac{(\lambda x)^n e^{-\lambda x}}{n!}, & x \geq 0 \end{cases}$$

Proof.

$$\begin{aligned}
F_k(x) &= \int_0^x \frac{\lambda^k t^{k-1} e^{-\lambda t}}{(k-1)!}\,\mathrm{d}t \\
&= \frac{\lambda^{k-1}}{(k-1)!}\left(\left(-t^{k-1} e^{-\lambda t}\right)\Big|_0^x + (k-1)\int_0^x t^{k-2} e^{-\lambda t}\,\mathrm{d}t\right) \\
&= -\frac{(\lambda x)^{k-1} e^{-\lambda x}}{(k-1)!} + \frac{\lambda^{k-1}}{(k-2)!}\int_0^x t^{k-2} e^{-\lambda t}\,\mathrm{d}t \\
&= -\frac{(\lambda x)^{k-1} e^{-\lambda x}}{(k-1)!} + F_{k-1}(x)
\end{aligned}$$

$$F_1(x) = 1 - e^{-\lambda x}$$

$$F_k(x) = 1 - \sum_{n=0}^{k-1} \frac{(\lambda x)^n e^{-\lambda x}}{n!}$$

□

- Mean:

$$E(X) = \frac{k}{\lambda}$$

- Variance:

$$\mathrm{Var}(X) = \frac{k}{\lambda^2}$$

- MGF:
$$M_X(t) = \left(1 - \frac{t}{\lambda}\right)^{-k}, \quad t < \lambda$$
- Mode:
$$\text{Mode}(X) = \frac{k-1}{\lambda}$$

  Proof.
$$f_X'(x) = \frac{(k-1-\lambda x)\lambda^k x^{k-2} e^{-\lambda x}}{(k-1)!}$$
$$f_X'(x) > 0 \iff x < \frac{k-1}{\lambda}$$
$$f_X'(x) < 0 \iff x > \frac{k-1}{\lambda}$$
□

- Scaling:
$$cX \sim \text{Exp}\left(k, \frac{\lambda}{c}\right), \quad c \in \mathbb{R}_{>0}$$
- PDF monotonicity: strictly increasing for $x \le \frac{k-1}{\lambda}$ and strictly decreasing for $x \ge \frac{k-1}{\lambda}$
- Relation to Poisson random variable: For $\lambda, x > 0$,
$$P(\text{Erlang}(k,\lambda) > x) = P(\text{Pois}(\lambda x) \le k-1),$$
  equivalently,
$$P(\text{Erlang}(k,\lambda) \le x) = P(\text{Pois}(\lambda x) \ge k.$$

  Proof.
$$\begin{aligned} P(\text{Erlang}(k,\lambda) > x) &= 1 - F_X(x) \\ &= \sum_{n=0}^{k-1} \frac{(\lambda x)^n e^{-\lambda x}}{n!} \end{aligned}$$
□

xi Geometric random variables converge in distribution to exponential random variables theorem

For any fixed $\lambda \in (0, \infty)$, let $\left(X_n \sim \text{Geom}\left(\frac{\lambda}{n}\right)\right)_{n\in\mathbb{N}}$, $\left(Y_n = \frac{X_n}{n}\right)_{n\in\mathbb{N}}$, and $Y \sim \text{Exp}(\lambda)$. Then
$$Y_n \xrightarrow{\text{d}} Y \text{ as } n \to \infty.$$

Proof.
$$F_{Y_n}(x) = 1 - \left(1 - \frac{\lambda}{n}\right)^{\lfloor nx \rfloor}$$
$$\lim_{n\to\infty} F_{Y_n}(x) = 1 - e^{-\lambda x}$$
□

xii Negative binomial random variables converge in distribution to Erlang random variables theorem

For any fixed $\lambda \in (0, \infty)$, let $\left(X_n \sim \text{NegBin}\left(r, \frac{\lambda}{n}\right)\right)_{n\in\mathbb{N}}$, $\left(Y_n = \frac{X_n}{n}\right)_{n\in\mathbb{N}}$, and $Y \sim \text{Erlang}(r, \lambda)$. Then

$$Y_n \xrightarrow{\text{d}} Y \text{ as } n \to \infty.$$

Proof. By decomposition to geometric and exponential random variables respectively and geometric random variables converge in distribution to exponential random variables theorem. □

XXVII Sample Mean, Bounds, and Convergences

i Sample mean of i.i.d. random variables

Let $\{X_i\}_{i=1}^n$ be a sequence of i.i.d. random variables and $X_i \sim X$ for all i. The sum of $X_1, X_2 \dots, X_n$ is

$$S_n = \sum_{i=1}^n X_i,$$

and the sample mean of $X_1, X_2 \dots, X_n$ is defined as

$$\bar{X} = \frac{S_n}{n},$$

which is a random variable that depends on $X_1, X_2 \dots, X_n$.

ii Mean and variance of sample mean

$$E(\bar{X}) = E(X).$$
$$\text{Var}\left(\bar{X}\right) = \frac{1}{n}\text{Var}(X).$$

Proof.

$$\begin{aligned}\text{Var}\left(\bar{X}\right) =& \text{Var}\left(\frac{1}{n}\sum_{i=1}^n X_i\right)\\ =&\frac{1}{n^2}\text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2}\left(\sum_{i=1}^n \text{Var}\left(X_i\right) + \sum_{i=1}^n\sum_{j=i+1}^n \text{Cov}\left(X_i, X_j\right)\right), \quad \text{Cov}\left(X_i, X_j\right) = 0\\ =&\frac{1}{n}\text{Var}(X)\end{aligned}$$

□

iii Markov's inequality

Let X be a non-negative random variable with finite mean. Then for any $a > 0$,

$$P(X \geq a) \leq \frac{E(X)}{a}.$$

Proof. Consider the derived random variable

$$Y = \begin{cases} a, & X \geq a, \\ 0, & \text{otherwise.} \end{cases}$$

$Y \leq X$ a.s., so $E(Y) = a \cdot P(X \geq a) \leq E(X)$. □

iv Chebyshev's inequality

Let X be a random variable with finite mean μ and finite variance σ^2. Then for any $\varepsilon > 0$,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Consider the derived random variable $Y = (X - \mu)^2$. $Y \geq 0$ a.s., so we can apply Markov's inequality to get

$$P(|X - \mu| \geq \varepsilon) = P(Y \geq \varepsilon^2) \leq \frac{E(Y)}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}.$$

□

Equivalently, for any $k > 0$,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

v Chernoff bound

Let X be a random variable. Then for $a \in \mathbb{R}$,

$$P(X \geq a) \leq \inf_{t \geq 0} e^{-at} M_X(t).$$

Proof. For any $t \geq 0$,

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov's inequality,

$$P(e^{tX} \geq e^{ta}) \leq \frac{E(e^{tX})}{e^{ta}} = e^{-at} M_X(t).$$

□

vi Tight

Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of random variables. It is called tight if

$$\lim_{x \to \infty} \left(\sup_n P(\left|X_n\right| > x) \right) = 0.$$

vii Lévy's continuity theorem or Lévy's convergence theorem

Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of random variables with characteristic functions $\{\varphi_n(t) = E\left(e^{itX_n}\right)\}_{n=1}^{\infty}$. If $\varphi_n(t)$ converges pointwise to some function $\varphi(t)$, then the following statements are equivalent

1. $\{X_n\}_{n=1}^{\infty}$ converges in distribution to some random variable X.

2. $\varphi(t)$ is a characteristic function of some random variable X.

3. $\{X_n\}_{n=1}^{\infty}$ is tight:

$$\lim_{x\to\infty}\left(\sup_n P(|X_n| > x)\right) = 0.$$

4. $\varphi(t)$ is continuous.

5. $\varphi(t)$ is continuous at $t = 0$.

And the X in the first statement has characteristic function $\varphi(t)$.

Proof. PLACEHOLDER □

viii Concentration inequality for sample mean

Let $\{X_i\}_{i=1}^{n}$ be a sequence of i.i.d. random variables with finite mean and finite variance, $X_i \sim X$ for all i, and $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Apply Chebyshev's inequality to $\bar{X}$, for any $\varepsilon > 0$,

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X})}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2},$$

or equivalently, for any $k > 0$,

$$P(|\bar{X} - \mu| \geq k\sigma) \leq \frac{1}{nkn^2}.$$

ix Weak law of large numbers (WLLN) or Khinchin's law

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with finite mean μ, $X_i \sim X$ for all i, and $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then

$$\bar{X}_n \xrightarrow{\mathrm{p}} \mu \text{ as } n \to \infty.$$

Proof. If $\mathrm{Var}(X) = \sigma^2$ is finite, we can prove by applying Chebyshev's inequality to $\bar{X}$, for any $\varepsilon, \delta > 0$, there exists $n \in \mathbb{N}$ such that

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq \delta.$$

By Taylor's theorem, the characteristic function of any random variable with finite mean μ can be written as

$$\varphi_X(t) = 1 + it\mu + o(t), \quad t \to 0.$$

$$\varphi_{\bar{X}_n}(t) = \left(\varphi_X\left(\frac{t}{n}\right)\right)^n = \left(1 + i\mu\frac{t}{n} + o\left(\frac{t}{n}\right)\right)^n \to e^{i\mu t} \text{ as } n \to \infty.$$

By Lévy' s continuity theorem,

$$\bar{X}_n \xrightarrow{\mathrm{d}} \mu \text{ as } n \to \infty.$$

Since μ is a constant

$$\bar{X}_n \xrightarrow{\mathrm{d}} \mu \text{ as } n \to \infty$$

is equivalent to

$$\bar{X}_n \xrightarrow{\text{p}} \mu \text{ as } n \to \infty.$$

□

x Strong law of large numbers (SLLN) or Kolmogorov's law

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with mean μ, $X_i \sim X$ for all i, $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$, and $E(|X|) < \infty$. Then

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu \text{ as } n \to \infty.$$

Proof. PLACEHOLDER □

xi (Lindeberg–Lévy or classical) central limit theorem (CLT)

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with finite mean μ and finite positive standard deviation σ, $X_i \sim X$ for all i, $S_n = \sum_{i=1}^{n} X_i$, and $\bar{X}_n = \frac{S_n}{n}$. Then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{\text{d}} \mathcal{N}(0,1) \text{ as } n \to \infty,$$

equivalently,

$$\frac{S_n - n\mu}{\sqrt{n}} \xrightarrow{\text{d}} \mathcal{N}(0,\sigma^2) \text{ as } n \to \infty,$$

equivalently,

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{\text{d}} \mathcal{N}(0,1) \text{ as } n \to \infty,$$

equivalently,

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{\text{d}} \mathcal{N}(0,\sigma^2) \text{ as } n \to \infty.$$

Proof. Define $Y_i = \dfrac{X_i - \mu}{\sigma} \sim Y$. Then

$$E[Y] = 0, \quad \text{Var}(Y) = 1$$

Define

$$Z_n = \frac{\sqrt{n}\left(\bar{X}_n - \mu\right)}{\sigma} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} Y_i$$

Let $\varphi(t) = E\left(e^{itY}\right)$ be the characteristic function of Y.

$$\varphi_{Z_n}(t) = \left(\varphi\left(\frac{t}{\sqrt{n}}\right)\right)^n$$

By Taylor's theorem,

$$\varphi(u) = 1 - \frac{u^2}{2} + o(u^2)$$

$$\varphi\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)$$
$$\varphi_{Z_n}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n$$
$$\lim_{n\to\infty} \varphi_{Z_n}(t) = e^{-\frac{t^2}{2}},$$

which is the characteristic function of $\mathcal{N}(0, 1)$.

By Lévy' s continuity theorem,

$$Z_n \xrightarrow{\text{d}} \mathcal{N}(0, 1).$$

$\square$

For large n, we often approx S_n with

$$S_n \approx \mathcal{N}(n\mu, n\sigma^2) \text{ as } n \to \infty.$$

xii Continuity correction

Continuity correction is an adjustment made when a discrete random variable is approximated using a continuous random variable.

XXVIII Approximate binomial random variable using CLT

Let

$$X_n \sim \text{Bin}(n, p).$$

Then

$$\frac{X_n - np}{\sqrt{np(1-p)}} \xrightarrow{\text{d}} \mathcal{N}(0, 1).$$

Proof. Suppose $\left(X_i \sim \text{Bern}(p)\right)_{i\in\mathbb{N}}$ is an i.i.d. sequence of random variables, which has mean p and variance $p(1-p)$. $X_n = \sum_{i=1}^{n} X_i$. By CLT,

$$\frac{X_n - np}{\sqrt{np(1-p)}} \xrightarrow{\text{d}} \mathcal{N}(0, 1).$$

$\square$

For large n,

$$X_n \approx \mathcal{N}(np, np(1-p))$$
$$P(X_n \le k) \approx \Phi\left(\frac{k + 0.5 - np}{\sqrt{np(1-p)}}\right)$$
$$P(X_n < k) \approx \Phi\left(\frac{k - 0.5 - np}{\sqrt{np(1-p)}}\right)$$
$$P(X_n = k) \approx \Phi\left(\frac{k + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - 0.5 - np}{\sqrt{np(1-p)}}\right)$$

XXIX Approximate Poisson random variable using CLT

Let

$$X_\lambda \sim \text{Pois}(\lambda).$$

Then

$$\frac{X_\lambda - \lambda}{\sqrt{\lambda}} \xrightarrow{\text{d}} \mathcal{N}(0,1) \quad \text{as} \quad \lambda \to \infty.$$

Proof. Suppose $\left(X_i \sim \text{Pois}\left(\frac{\lambda}{\lfloor\lambda\rfloor}\right)\right)_{i\in\mathbb{N}}$ is an i.i.d. sequence of random variables, which has mean $\frac{\lambda}{\lfloor\lambda\rfloor}$ and variance $\frac{\lambda}{\lfloor\lambda\rfloor}$. $X_\lambda = \sum_{i=1}^{\lfloor\lambda\rfloor} X_i$. By CLT,

$$\frac{X_\lambda - \lambda}{\sqrt{\lambda}} \xrightarrow{\text{d}} \mathcal{N}(0,1).$$

□

For large λ,

$$X_\lambda \approx \mathcal{N}(\lambda, \lambda)$$

$$P(X_\lambda \le k) \approx \Phi\left(\frac{k + 0.5 - \lambda}{\sqrt{\lambda}}\right)$$

$$P(X_\lambda < k) \approx \Phi\left(\frac{k - 0.5 - \lambda}{\sqrt{\lambda}}\right)$$

$$P(X_\lambda = k) \approx \Phi\left(\frac{k + 0.5 - \lambda}{\sqrt{\lambda}}\right) - \Phi\left(\frac{k - 0.5 - \lambda}{\sqrt{\lambda}}\right)$$

XXX Approximate negative binomial random variable using CLT

Let

$$X_r \sim \text{NegBin}(r, p).$$

Then

$$\frac{X_r - \frac{r}{p}}{\sqrt{r\frac{1-p}{p^2}}} = \frac{pX_r - r}{\sqrt{r(1-p)}} \xrightarrow{\text{d}} \mathcal{N}(0,1).$$

Proof. Suppose $\left(X_i \sim \text{Geom}(p)\right)_{i\in\mathbb{N}}$ is an i.i.d. sequence of random variables, which has mean $\frac{1}{p}$ and variance $\frac{1-p}{p^2}$. $X_r = \sum_{i=1}^{r} X_i$. By CLT,

$$\frac{X_r - \frac{r}{p}}{\sqrt{r\frac{1-p}{p^2}}} = \frac{pX_r - r}{\sqrt{r(1-p)}} \xrightarrow{\text{d}} \mathcal{N}(0,1).$$

□

For large r,

$$X_r \approx \mathcal{N}(\frac{r(1-p)}{p}, \frac{r(1-p)}{p^2})$$

$$P(X_r \le k) \approx \Phi\left(\frac{k+0.5-\frac{r(1-p)}{p}}{\sqrt{\frac{r(1-p)}{p^2}}}\right)$$

$$P(X_r < k) \approx \Phi\left(\frac{k-0.5-\frac{r(1-p)}{p}}{\sqrt{\frac{r(1-p)}{p^2}}}\right)$$

$$P(X_r = k) \approx \Phi\left(\frac{k+0.5-\frac{r(1-p)}{p}}{\sqrt{\frac{r(1-p)}{p^2}}}\right) - \Phi\left(\frac{k-0.5-\frac{r(1-p)}{p}}{\sqrt{\frac{r(1-p)}{p^2}}}\right)$$

XXXI Statistical Inference

i Statistical inference

For some parameter space Θ, we can estimate some properties $t(\theta)$, i.e., a function of $\theta \in \Theta$, by sampling some (probability) distribution or statistical model:

- Data generation: Collects samples. A typical scenario of the data is a sequence of samples drawn i.i.d. from the same distribution $X \sim P_\theta$, where P_θ is the PDF of absolutely continuous X or PMF of discrete X.
- Estimate via an inference algorithm (also called (decision) rule): Compute a function φ of the observed data x which gives an estimate of $t(\theta)$, called estimator and denoted $\hat{\theta}$. The algorithm may be deterministic, i.e., purely a function of X, or randomized, i.e., $\varphi(X, U)$ also depends on some random seed U.
- Performance evaluation: We measure the performance by some loss function $\ell(\varphi(X, U), t(\theta))$. We want to design an inference algorithm that minimizes the expected loss (also called the risk) $R(\varphi; \theta) = E_{X\sim P_\theta, U}(\ell(\varphi(X, U), t(\theta)))$.

ii Parameter estimation

Parameter estimation is the estimation of the parameters of a probability distribution or statistical model. The target parameters to be estimated may determine the distribution or partially determine it. We may estimate mean, variance, probability of an event, etc.

Two main scenarios of parameter estimation are point estimation and interval estimation.

iii Point estimation

In point estimation, we estimate the target parameter θ itself. The estimator is called point estimator.

iv Interval estimation

In interval estimation, we estimate an interval that contains the target parameter θ with high probability. The estimator is called interval estimator.

v Determine the best inference algorithm

A goal of statistical inference is to determine the best inference algorithm that mimizes the risk for a given loss function and parameter space, i.e., find φ^* such that

$$R(\varphi^*;\theta) \leq R(\varphi;\theta)$$

for all $\theta \in \Theta$ for all inference algorithms φ.

However, this is not always possible.

There are two main paradigms for statistical inference: the minimax paradigm and Bayesian paradigm.

vi Minimax paradigm

In the minimax paradism, the risk $R(\varphi)$ of an inference algorithm φ is defined as the worst-case expected loss over all possible values of θ, called worst-case risk, that is,

$$R(\varphi) = \max_{\theta\in\Theta} R(\varphi;\theta).$$

The minimax risk is defined as

$$R^* = \min_{\varphi} R(\varphi),$$

and the minimax optimal rule is defined as

$$\varphi^* = \arg_{\varphi} \min_{\varphi} R(\varphi).$$

vii Bayesian paradigm

In Bayesian paradigm, we incorporate prior knowledge about the parameter and treat the unknown parameter θ as a random variable with a prior distribution $\pi(\theta)$, and the risk $R(\varphi)$ of an inference algorithm φ is defined as the expected loss with respect to the prior distribution, called average-case risk, that is,

$$R(\varphi) = E_{\theta\sim\pi} R(\varphi;\theta).$$

The Bayes risk is defined as

$$R^* = \min_{\varphi} R(\varphi),$$

and the Bayes optimal rule is defined as

$$\varphi^* = \arg_{\varphi} \min_{\varphi} R(\varphi).$$

For discrete prior random variable, $\pi(\theta)$ is its PMF and point estimation is used. For absolutely continuous prior random variable, $\pi(\theta)$ is its PDF and interval estimation is used.

viii Hypothesis testing

There are m hypotheses $H_0, H_1, \dots, H_{m-1}$

$$H_\theta : X \sim P_\theta, \quad \theta \in \{0, 1, \dots, m-1\},$$

and we have to determine which hypothesis is true based on the observed data $x \in \mathrm{X}$.

The goal is to design a (decision) rule $\varphi : \mathrm{X} \to \{0, 1, \dots, m-1\}$ that maps the observed data to a hypothesis, so that a certain risk is minimized.

A common loss function is the 0-1 loss:

$$\ell(\varphi(X), \theta) = \begin{cases} 0, & \varphi(X) = \theta \\ 1, & \varphi(X) \neq \theta \end{cases}$$

The risk is

$$R(\varphi; \theta) = P_{X \sim P_\theta}(\varphi \neq \theta).$$

ix Best Bayesian decision rule or Maximum a posteriori (MAP) decision rule

If we have a prior discrete distribution $\pi(\theta)$ over the hypothesis, the Bayesian optimal rule φ^* that minimized the average-case risk $E_{\theta \sim \pi} R(\varphi; \theta)$ is, for every $x \in \mathrm{X}$,

$$\varphi^*(x) = \arg_\theta \max_\theta P_\theta(x)\pi(\theta).$$

Proof. Recall that minimize the risk is equivalent to maximizing $P_{X,\Theta}(\varphi(X) = \Theta)$.

For discrete X,

$$\max_\varphi E_{X,\Theta}\left(\mathbf{1}_{\{\varphi(X)=\Theta\}}\right) = \max_\varphi \sum_x \sum_\theta \mathbf{1}_{\{\varphi(X)=\theta\}} P_\theta(x)\pi(\theta) = \sum_x \max_\theta P_\theta(x)\pi(\theta)$$

and

$$\varphi(x) = \arg_\theta \max_\theta P_\theta(x)\pi(\theta).$$

For absolutely continuous X,

$$\max_\varphi E_{X,\Theta}\left(\mathbf{1}_{\{\varphi(X)=\Theta\}}\right) = \max_\varphi \int_{-\infty}^{\infty} \sum_\theta \mathbf{1}_{\{\varphi(X)=\theta\}} P_\theta(x)\pi(\theta)\,\mathrm{d}x = \int_{-\infty}^{\infty} \max_\theta P_\theta(x)\pi(\theta)\,\mathrm{d}x$$

and

$$\varphi(x) = \arg_\theta \max_\theta P_\theta(x)\pi(\theta).$$

□

For discrete X,

$$P_\theta(x)\pi(\theta) = p_{X,\Theta}(x, \theta) = p_X(x)p_{\Theta|X}(\theta|x).$$

For continuous X,

$$P_\theta(x)\pi(\theta) = \frac{\mathrm{d}}{\mathrm{d}x}F_{X,\Theta}(x, \theta) = f_X(x)p_{\Theta|X}(\theta|x).$$

$p_X(x)$ and $f_X(x)$ is not dependent on θ, and $p_{\Theta|X}(\theta|x)$ is the posterior distribution of Θ given $X = x$. Thus the rule can be rewritten as

$$\varphi^*(x) = \arg_\theta \max_\theta p_{\Theta|X}(\theta|x).$$

This is called the maximum a posteriori (MAP) (decision) rule, which chooses the hypothesis with highest posterior probability given data x.

x Maximum likelihood (ML) decision rule

In the case of $\pi(\theta) = \frac{1}{m}$ for all θ, the MAP decision rule reduces to the maximum likelihood (ML) decision rule:

$$\varphi^*(x) = \arg_\theta \max_\theta P_\theta(x) = \arg_\theta \max_\theta L(\theta|x),$$

where $L(\theta|x): P_\theta(x)$ is called likelihood function.

xi Binary hypothesis testing

In the special case of binary hypothesis testing, there are only two hypotheses $H_0 : X \sim P_0$ and $H_1 : X \sim P_1$, called null hypothesis and alternative hypothesis respectively.

The likelihood ratio is

$$L(x) = \frac{P_1(x)}{P_0(x)}$$

and the MAP decision rule is

$$\varphi^*(x) = \begin{cases} 1, & L(x) > \frac{\pi(1)}{\pi(0)} \\ 0, & L(x) \leq \frac{\pi(1)}{\pi(0)}. \end{cases}$$

If the goal is the 0-1 loss, there are two kinds of errors:

- Type I error, false positive, or false alarm: Rejecting H_0 when H_0 is true, which occurs with probability $R(\varphi; 0) = P_{X \sim P_0}(\varphi(X) = 1)$, called false alarm probability and denoted P_{FA}.
- Type II error, false positive, or missing: Accepting H_0 when H_1 is true, which occurs with probability $R(\varphi; 1) = P_{X \sim P_1}(\varphi(X) = 0)$, called missing probability and denoted P_{MISS}.

Total error probability is

$$P_{\text{ERR}} = P_{\text{FA}}\pi(0) + P_{\text{MISS}}\pi(1).$$

xii Bayesian-optimal test for binary hypothesis testing

$$R(\varphi) = E_{\theta \sim \pi} R(\varphi; \theta) = \pi(0)R(\varphi; 0) + \pi(1)R(\varphi; 1).$$

The MAP rule φ^* can be shown to be the likelihood ratio rule, for everyn$x \in \mathrm{X}$,

$$\varphi^*(x) = \begin{cases} 1, & \frac{P_1(x)}{P_0(x)} > \frac{\pi(0)}{\pi(1)} \\ 0, & \text{otherwise} \end{cases}$$

The ratio $\frac{P_1(x)}{P_0(x)}$ is called the likelihood ratio, which measures how much more likely the observed data x is under H_1 compared to H_0.

xiii Mean squared error (MSE) of point estimators

The loss function by collecting data x can be squared error loss:

$$\ell(\varphi(x), t(\theta)) = |\varphi(x) - \theta|^2 .$$

Then

$$R(\varphi; \theta) = \mathrm{MSE}\left(\hat{\theta}\right) = E_{X\sim P_\theta}((\varphi(x) - \theta)^2).$$

xiv Bias of point estimators

The bias of $\varphi(X)$ is defined as

$$\mathrm{Bias}(\varphi(X)) = E(\varphi(X)) - t(\theta).$$

The estimator is called unbiased if $\mathrm{Bias}(\varphi(X)) = 0$ for all θ.

xv Weak consistency of point estimators

An estimator $\varphi(X)$ is weakly consistent if $\varphi(X_1, \dots, X_n) \xrightarrow{P} t(\theta)$ as $n \to \infty$ for all θ.

xvi Point estimation of variance

If the samples $X_1, \dots, X_n \sim P_\theta$ where $\theta = (\mu, \sigma^2)$ and μ is known, then we can estimate σ^2 using the estimator

$$\varphi(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^{n} \left(X_i - \mu\right)^2 .$$

The estimator is unbiased, since $E(\varphi(X_1, \dots, X_n)) = \sigma^2$.

If μ is unknown, we may use sample mean $\bar{X}$ to estimate μ and get the estimator for variance

$$\varphi(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2 .$$

However, this estimator is biased, since

$$E(\varphi(X_1, \dots, X_n)) = \frac{n-1}{n}\sigma^2 \neq \sigma^2.$$

Proof.

$$X_i - \bar{X} = \frac{n-1}{n} X_i + \frac{1}{n} \sum_{j \neq i} X_j$$

$$E((X_i - \bar{X})^2) = \left(\frac{n-1}{n}\right)^2 \sigma^2 + \frac{n-1}{n^2}\sigma^2 = \frac{n-2}{n}\sigma^2.$$

□

The estimator is weakly consistent since $\varphi(X_1, \dots, X_n) \xrightarrow{P} \sigma^2$ as $n \to \infty$.

To get an unbiased estimator of σ^2, we can multiply the previous estimator by $\dfrac{n}{n-1}$ as

$$\bar{V}_n = \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2 ,$$

called sample variance.

Hence, the sample mean $\bar{X}_n$ and sample variance $\bar{V}_n$ are unbiased and weakly consistent estimators of the population mean μ and variance σ^2 respectively.

xvii Estimation of the probability of an event

Estimating the probability $P(A)$ of an event A is just estimating the mean of a Bernoulli variable $Y \sim \text{Bern}(P(A))$. We can just use the sample mean of Y to estimate $P(A)$, i.e.,

$$\bar{P}_n(A) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_A(X_i).$$

It's an unbiased and weakly consistent estimator of $P(A)$ since $E(\bar{P}_n(A)) = P(A)$ and $\bar{P}_n(A) \xrightarrow{P} P(A)$ as $n \to \infty$.

xviii Non-asymptotic performance

Statistical risks, either in the minimax or Bayesian sense, can be used to quatify the non-asymptotic performance of an estimator, that is, we may want to know how fast the convergence

$$\lim_{n\to\infty} P(\left|\varphi(X_1, \dots, X_n) - t(\theta)\right| > \varepsilon) = 0$$

is, that is, find the minimum number of samples n such that

$$P(\left|\varphi(X_1, \dots, X_n) - t(\theta)\right| > \varepsilon) \le \delta$$

for each δ.

xix 0-1 loss of interval estimator

The loss function by collecting data x can be the 0-1 loss:

$$\ell(\varphi(x), t(\theta)) = \begin{cases} 0, & \theta \in \varphi(x) \\ 1, & \theta \notin \varphi(x) \end{cases}$$

Then

$$R(\varphi; \theta) = P_{X\sim P_\theta}(\theta \notin \varphi(X)).$$

xx Coverage probability and confidence coefficient of interval estimator

The coverage probability of an interval estimator $I(X) = [L(X), U(X)]$ for a parameter θ is defined as $P(\theta \in I(X))$ whre the probability is taken over the distribution of the data X.

The confidence coefficient of the interval estimator is the minimum of the coverage probability over all possible values of θ, i.e., $\min_\theta P(\theta \in I(X))$.

An estimated interval with a confidence coefficient is also called confidence interval.

For an interval estimator, there are two key performance factors that quantify it,

- Coverage probability.
- Interval width: the length of the interval, which reflects the precision of the estimation.

xxi Interval estimation of the population mean

Given an interval width parameter ε and a confidence level $1-\delta$, our goal is to find an interval of width 2ε such that the population mean μ is contained in the interval with probability at least $1-\delta$.

The interval can be found by taking samples $X_1, \ldots, X_n$ from the distribution and computing sample mean $\bar{X}_n$. Then the interval can be constructed as $[\bar{X}_n - \varepsilon, \bar{X}_n + \varepsilon]$.

xxii Chebyshev's inequality on confidence coefficient of interval estimator of the population mean

If the distribution has a upper bound of variance $\sigma^2 \le b$, then by Chebyshev's inequality, we have

$$P(|\bar{X}_n - \mu| > \varepsilon) \le \frac{b}{n\varepsilon^2}.$$

Hence, for any $\delta > 0$,

$$n \ge \frac{b}{\delta\varepsilon^2} \implies P(|\bar{X}_n - \mu| > \varepsilon) \le \delta.$$

And the confidence coefficient of the interval estimator for this case is at least

$$1 - \frac{b}{\delta\varepsilon^2}.$$

xxiii Interval estimator of Guassian distribution

If the distribution is Guassian distribution with unknown mean μ and variance, then we can use the sample variance $\bar{V}_n$ to perform interval estimation.

In this case, $\frac{\bar{X}_n - \mu}{\sqrt{\frac{\bar{V}_n}{n}}}$ follows the Student's t-distribution with $n-1$ degrees of freedom:

$$T := \frac{\bar{X}_n - \mu}{\sqrt{\frac{\bar{V}_n}{n}}} \sim t_{n-1}.$$

Then, for any $\delta > 0$, there exists a constant t^* such that $P(T > t^*) = \frac{\delta}{2}$. Then the confidence interval with confidence coefficient $1-\delta$ is

$$\left[\bar{X}_n - t^*\sqrt{\frac{\bar{V}_n}{n}}, \bar{X}_n + t^*\sqrt{\frac{\bar{V}_n}{n}}\right].$$